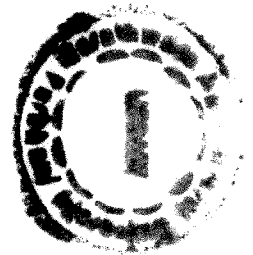


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**“STUDIES ON DRY ROOT ROT OF CHICKPEA
(*Cicer arietinum* L.) CAUSED BY *Macrophomina
phaseolina*.(Tassi) Goid.”**

By
MISS. SAYYAD SURAIYYANAJ ISSAK
B.Sc. (Agri.)

T 6244



**MASTER OF SCIENCE
(Agriculture)
In
PLANT PATHOLOGY**

**DEPARTMENT OF PLANT PATHOLOGY
COLLEGE OF AGRICULTURE, LATUR
MARATHWADA KRISHI VIDYAPEETH,
PARBHANI
2011**

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DISSERTATION

Submitted to the

MARATHWADA KRISHI VIDYAPEETH, Parbhani

*in partial fulfillment of the requirement for the
degree of*

MASTER OF SCIENCE

(Agriculture)

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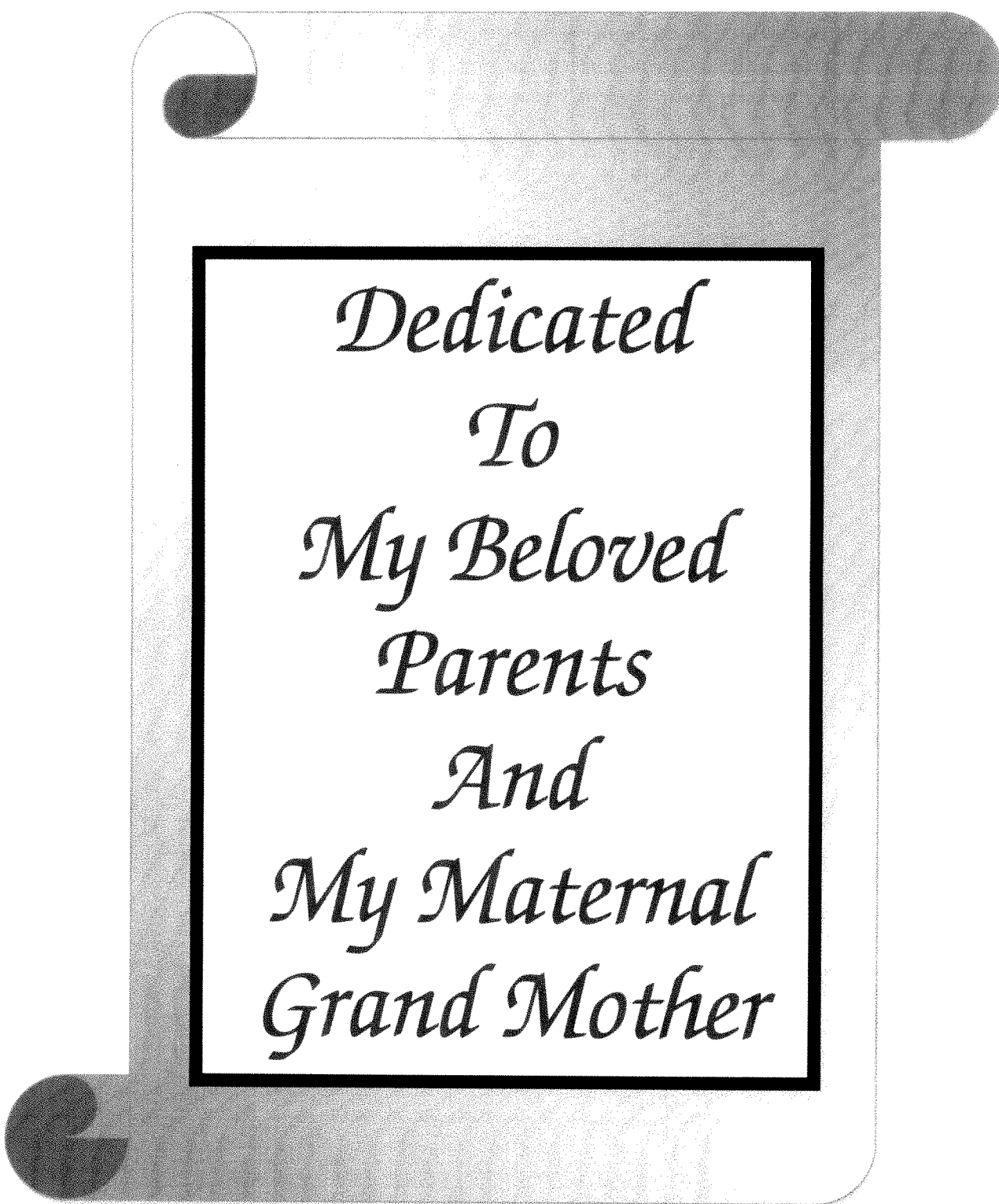
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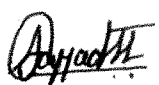
*Dedicated
To
My Beloved
Parents
And
My Maternal
Grand Mother*

CANDIDATE'S DECLARATION

*I hereby declare that the dissertation
or part there of has not been
previously submitted by me
for a degree of any
University or
Institute*

Place : LATUR

Date : 31 / 3 / 2011


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CERTIFICATE I

This is to certify that the dissertation entitled “**STUDIES ON DRY ROOT ROT OF CHICKPEA (*Cicer arietinum* L.) CAUSED *Macrophomina phaseolina*.(Tassi) Goid**” submitted by **MISS. SAYYAD SURAIYYANAJ ISSAK** to the Marathwada Krishi Vidyapeeth, Parbhani in partial fulfillment of the requirements for the degree of **MASTER OF SCIENCE (Agriculture)** in the subject of **PLANT PATHOLOGY** is record of original and bonafide research work carried out by her under my guidance and supervision. It is of sufficiently high standard to warrant it's presentation for the said degree.

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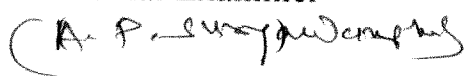
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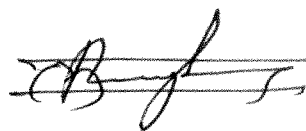
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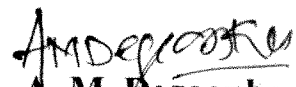
This is to certify that the dissertation entitled “**STUDIES ON DRY ROOT ROT OF CHICKPEA (*Cicer arietinum* L.) CAUSED *Macrophomina phaseolina*.(Tassi) Goid**” submitted by **MISS. SAYYAD SURAIYYANAJ ISSAK** to the Marathwada Krishi Vidyapeeth, Parbhani in partial fulfillment of the requirement for the degree of **MASTER OF SCIENCE (Agriculture)** in the subject of **PLANT PATHOLOGY** has been approved by the student’s advisory committee after oral examination in collaboration with the External Examiner.


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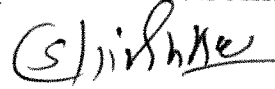

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(Miss. Sayyad S.I.)

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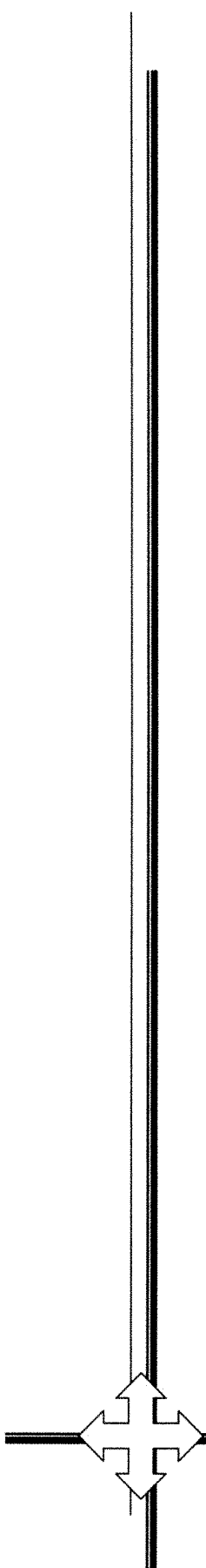
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INTRODUCTION



Chapter I

INTRODUCTION

Chickpea (*Cicer arietinum* L.) is the world's third most important crop, after dry Beans (*Phaseolous vulgaris* L.) and dry Peas (*Pisum sativum* L.) (Vishwadhar and Gurha, 1998). Although, Chickpea is predominantly consumed as pulse, dry Chickpea is also used in preparing a variety of snack foods, sweets and condiments, and green fresh Chickpeas are commonly consumed as vegetable.

Nutritionally, Chickpea is relatively free from various antinutritional factors having higher protein digestibility and rich in phosphorus and calcium than the other pulses (Swaminathan, 1989). Because of its higher fat content and better fibre digestibility, Chickpea hold great promise as protein calorie. Chickpea are healthy food as they contain many nutritional elements that are deemed necessary for growth of our body. Chickpea contains vital vitamin-B₂, vitamin-B₃, vitamin-C, vitamin-D, vitamin-E, vitamin-K, folic and pantothenic acid. They also contain many important minerals like calcium, iron, molybdenum, potassium, manganese, copper, zinc etc. Other nutrients that are found in Chickpeas are dietary fibres, carbohydrates (61.5 per cent), proteins (25.29 per cent), fats (4.5 per cent), water *etc.*

Chickpea is mainly used for human consumption as well as animal feed both for ruminants and non-ruminants. It is consumed as whole seed, Dal, fried, boiled, salted or more generally which is cooked. Chickpea flour is mixed with wheat flour to improve protein content of wheat flour and is used in making missi roti. The flour of dehusked Chickpea called 'Besan' is widely used in making Pakodas, Kadhi, Namkeens and several snack foods. Exudation of leaves locally called 'Amb' contain malic and oxalic acid which posses medicinal value for bronchitis, cholera, constipation, diarrhea, digestive disorder, Snake bites,

warts and blood purification, and germinated seeds are recommended to cure scurvy diseases. Chickpea straw has a forage value comparable to other straw commonly used as livestock feed. Because of this diversified use of the crop and its ability to grow better which low input under harsh edaphic and arid environment than any other crops, it is an important component of the cropping system and subsistence farming in the Indian subcontinent.

Chickpea is major pulse crop in India and accounts for 40-45 per cent of total pulse production. India is the largest producer in world which accounts 66 per cent of total world production. Different Indian states have different name for Chickpea like it is called as Chana in northern states, Chhole in Punjab, Kadala in Tamilnadu, Kadala in Kerala.

In India, the total area under Chickpea during 2007-2008 was 7.58 million ha with the production of 5.9 million tonnes and average productivity 780 kg/ha (Anonymous, 2008). Where as in Maharashtra, the total area under Chickpea during 2007-2008 was 13.53 lakh ha with the production of 10.04 lakh tonne and average productivity of 740 kg/ha, which is low as compared to national average (Anonymous, 2008).

Chickpea, which occupies third position in total area under cultivation and considered as poor man's crop but the productivity of crops is very low and needs concentrated efforts in order to bridge the gap between requirement and production in pulses. Several factors have been reported to be responsible for low productivity of Chickpea and amongst which dry root rot disease plays an important role. Fifty different pathogens have so far been reported on Chickpea (Nene *et al.* 1989) including diseases caused by fungal, bacterial, nematodes, mycoplasmal and viral pathogens. Several diseases reported from all growing states in India viz. Punjab, Delhi, Madhya Pradesh, Bangladesh, Bihar, Tamilnadu,

Uttar Pradesh, Karnataka, Gujarat, Maharashtra and U.S.A. (Nene *et al.* 1989).

A large number of diseases have been reported on Chickpea right from early seedling stage up to maturity; of which wilt (*Fusarium sp.*), dry root rot (*Macrophomina phaseolina* (Tassi) Goid.), wet root rot (*Rhizoctonia solani*), foot rot (*Operculella padwickii*), stem rot (*Sclerotiana sclerotiorum*), blight (*Ascochyta rabie*), stunt and mosaic (virus) are important. Amongst these entire diseases root rots such as dry root rot (*Macrophomina phaseolina*), black root rot (*Fusarium solani*) and wet root rot (*Rhizoctonia solani*) have been reported to cause severe losses from seedling to maturity stages of crop (Mc Rae, 1932). Chickpea root rot; collar rot and wilt were the most serious diseases which caused severe yield losses of 30-60 per cent under favourable conditions. (Nene, 1985). Quantitative and qualitative yield losses (about 30-70%) incurred by root rot (*Macrophomina phaseolina*) disease in Chickpea as well as in other crops like Groundnut, Blackgram, Greengram, Clusterbean, Sunflower, Safflower, Sesamum etc. were reported earlier by several workers. (Nene, 1985; Ahmed and Mohammad, 1986). Dry root rot caused by *Macrophomina phaseolina* was one of the major production constraints that caused 10-20 per cent annual losses in Chickpea (Vishwadhar and Chaudhary, 2001).

The *Macrophomina phaseolina* is one of the major soil born pathogen, facultative saprophyte and can survive in soil several year in the absence of susceptible host and commonly cause root rot disease (Haware *et al.* 1996). Considering the nature of damage and survival ability of the fungus, use of resistant varieties is the only economical and practical solution.

Research efforts had already been directed in this direction at ICRISAT, Hyderabad and at all India Co-ordinate Research Projects of,

Pulses. A number of dry root rot resistant varieties have been identified and released for large scale cultivation (Pande *et al.* 2004, and Reddy *et al.* 1990).

In recent past, few management strategies for combating these diseases have been suggested. These methods include use of resistant or tolerant cultivars, crop rotation, effective preparatory tillage operations and seed treatment with fungicides (Shughas *et al.*, 1995). However, these practices have limited success. High yielding, good quality, resistant varieties for majority of Chickpea growing areas in India are not available. Crop rotations preparatory tillage operations are beneficial on long term basis and helpful in partial management of the diseases. Chemical management is neither economical nor eco-friendly and thus it is not practicable. In light situation there is an acute need to develop integrated disease management module with all the efficient and economical attributes.

Considering importance of the disease in the farmer's field, and yield losses caused, the present investigations were undertaken to find the major causal organisms involved in Chickpea dry root rot in Marathwada region of Maharashtra state for formulation of the integrated disease management strategies with following objectives.

OBJECTIVES:

1. Survey on dry root rot of Chickpea in Latur district.
2. To prove the pathogenicity of *Macrophomina phaseolina* and cultural characteristics of the pathogen.
3. Evaluation of fungicides, bioagents and organic amendments against *Macrophomina phaseolina* in(Pot culture experiment).
4. Evaluation of fungicides and bioagents in *in vivo* against *Macrophomina phaseolina*.
5. Screening of Chickpea varieties, germplasm line and genotypes against *Macrophomina phaseolina*.

*REVIEW
OF
LITERATURE*



Chapter II

REVIEW OF LITERATURE

2.1 Occurance

Uppal (1931), reported that *Rhizoctonia bataticola* (*M. phaseolina*) for the first time from India on Jowar, which caused seedling blight and hallow stem.

Agrawal and Sarbhoy (1976), reported that widespread occurrence of *Rhizoctonia bataticola* (*M. phaseolina*), causing charcoal rot of Soybean and most of the country the pathogen had been found to infect the crop in almost all stages of crop growth.

Saksena (1979), reported that *Rhizoctonia bataticola* (*M. phaseolina*) was the widespread and destructive plant pathogen. The pathogen had been considered of great economic importance because earlier, the pathogens were reported to cause root rot, collar rot, and affect various aerial plant parts and causes stalk rot, leaf spot and blights.

Vishwadhar and Sarbhoy (1989), reported that *Rhizoctonia bataticola* (Taub), Butler (Pycnidial stage, *M. phaseolina*) caused seedling rot, root rot, ashy stem blight and charcoal rot in the crops like Soybean, Mung bean, Sunflower, Urdbean *etc.* in India.

2.2 Yield losses

Quantitative and qualitative yield losses incurred by rot (*Macrophomina phaseolina* (Tassi) Goid.) disease in Chickpea as well as in other crops like Groundnut, Blackgram, Greengram, Clusterbean, Sunflower, Safflower, Sesamum *etc.* were reported earlier by several workers.

Nene (1985), reported that Chickpea root rot, collar rot and wilt were the most serious diseases which caused severe yield losses of 30-60 per cent under favourable conditions.

Queasier and Mohammad (1986), reported that yield losses due to *Rhizoctonia* root rot of Chickpea in Bihar state and observed that the disease caused about 50 to 71 per cent yield losses in Chickpea.

Sen (2000), reported that *Macrophomina phaseolina* affected a variety of field crops and vegetable crops besides a wide range of Urdbean plant species and caused charcoal rot, which accounted for crop losses to the tune of 10 to 20 per cent.

Vishwadhar and Chaudhary (2001), reported that Chickpea was susceptible to several fungal diseases amongst them, dry root rot caused by *Macrophomina phaseolina* was one of the major production constrains that caused 10-20 per cent annual losses.

2.3 Pathogen and symptomology of the disease

Narashimhan (1928), reported that examination of wilted plants of Chickpea received in January 1928 from Bellary district of Karnataka and Coimbatore (Tamilnadu) revealed that some species were associated with *Rhizoctonia spp.* and some with *Fusarium spp.* This appears to be the first report on wilt of Chickpea.

Westerland *et al.* (1994), reported that disease of Chickpea from the Central coastal areas of California, caused by at least five fungi one of them was *Macrophomina phaseolina*.

Nene *et al.* (1978), found continuous black discoloration of pith and xylem vessels of roots and basal shoots as one of the main characteristics symptoms of wilt in Chickpea induced by *Rhizoctonia bataticola* (*M. phaseolina*).

Biswas and Gupta (1981), described that symptoms induced on Chickpea by isolates of *Rhizoctonia bataticola* (*M. phaseolina*),

Singh *et al.* (1982), reported that mycelia inoculums of *Rhizoctonia bataticola* (*M. phaseolina*) was more effective than sclerotia in reducing germination and increased the percentage of infected parts of Chickpea.

Tripathi and Sharma (1983), reported that the incidence of *Rhizoctonia bataticola* (*M. phaseolina*) in southern Haryana states was high in late October to mid November and decreased in December and January and then increased again February and March.

Taya *et al.* (1988), reported that incidence of *Rhizoctonia bataticola* (*M. phaseolina*) causing dry root rot of Chickpea was higher in sandy loamy soil than clay black soils while low moisture was conducive to disease development.

Singh *et al.* (1990), observed that *Rhizoctonia bataticola* (*M. phaseolina*) inoculated roots of Chickpea upon microscopic examination showed disintegration of cortical tissues and in addition mycelial and sclerotial bodies of *Rhizoctonia bataticola* (*M. phaseolina*) plugging xylem vessels were observed in longitudinal sections.

Beniwal *et al.* (1992), reported that out of five important diseases of Chickpea, wilt and dry root rot are most important. They also described the diagnostic symptoms of each of these diseases.

2.4 Survey on dry root rot of Chickpea in Latur district

Sharma *et al.* (1983), conducted on survey of wilt and root rot incidence of *Cicer arietinum* in northern region of Madhya Pradesh. Survey was conducted on Chickpea growing areas of 30 villages of northern region of Madhya Pradesh, wilt caused by *Fusarium sp.* and dry root rot caused by (*Macrophomina phaseolina*). The average of wilt incidence ranged from 14.90 to 37.99 per cent and root rot from 3.58 to 20.63 per cent.

Pandey and Singh (1990), in a survey in Allahabad region, India observed Chickpea to be infected by *F.oxysporum* and *R.bataticola*.

Singh *et al.* (1991), conducted survey on prevalence and severity of dry root rot of Sesamum caused by *M. phaseolina* in the different states Delhi, Haryana and Uttar Pradesh during 1980-1982. The incidence of root rot of Sesamum in cultivated field varies from 60.0 to 71.50 percent depending upon soil condition and crop season. It also showed that the average of 17.01 per cent root rot is present in all the area surveyed.

Srivastava *et al.* (2002), were conducted survey in the Vidhyan plateau zone of Madhya Pradesh, during 1998-99 to evaluate the prevalence of wilt and dry root rot diseases in Chickpea.

Abbas *et al.* (2003), conducted field survey to determine the incidence and economic damage caused by charcoal rot in Soybean and reported that 4.8 to 8.0 per cent incidence of the disease.

Prajapati *et al.* (2003), conducted survey on incidence of Chickpea dry root rot in Uttar Pradesh and Uttaranchal and reported that the incidence of dry root rot of Chickpea caused by (*Macrophomina phaseolina*) in 20 locations in Uttar Pradesh and Uttaranchal.

Prajapati *et al.* (2004), conducted survey on incidence of Chickpea dry root rot in Uttar Pradesh. Survey was conducted at 20 locations of different agro-climatic regions of Uttar Pradesh. The disease was wide spread in occurrence in UP with 19.0 to 42.0 per cent incidence. (Average, 29.45 per cent).

Rana *et al.* (2009), conducted survey on Pea diseases and identification of fungi associated with and root rot complex in Himachal Pradesh. Survey was conducted the disease were prevalent in all Pea growing areas of the state during 2006-2008. The incidence of wilt and root rot was 20 per cent in the temperate wet, 15 per cent in sub temperate, 12 per cent in the temperate dry and 11 per cent in subtropical regions.

2.5 Collection, isolation and pathogenicity

2.5.1 Isolation and pathogenicity

Bhate (1983), isolated (*Macrophomina phaseolina*) from roots of Green gram and proved its pathogenicity.

Bhatti *et al.* (1987) isolated the pathogen viz. *Fusarium spp.* (42 per cent), *R. solani* (28 per cent), *Macrophomina phaseolina* (17 per cent) and *Verticillium alboatrum* (4 per cent) from wilted *Cicer arietinum* plants. From these the first 4 caused 100 per cent root rot in artificially sick soil.

Singh and Nene (1990), isolated pure cultures of *Rhizoctonia bataticola* (*M. phaseolina*) from infected Chickpea, pigeonpea, Groundnut and Sorghum and reported that Chickpea isolate was more pathogenic to Safflower, Sorghum, Bajra and Pigeonpea than to Chickpea and further suggested that the Chickpea isolate was the most aggressive.

Vishwadhar and Sarbhoy (1991), studied on a typical isolates of *R. bataticola* (*M. phaseolina*) reported that, *R. bataticola* (*M. phaseolina*) was isolated from pods and seeds of few diseased Soybean plant.

Devi and Singh (1998), studied on virulence of *Macrophomina phaseolina* isolates from Blackgram and Green gram and reported that, fifty six *Macrophomina phaseolina* isolates were obtained from roots, stem, pods and seeds of Blackgram and Greengram from 11 different localities of North, South, and Central India. All isolates were categorized into different groups.

Patel and Anahosur (2000), found association frequency of soil born fungi with wilted Chickpea plants. Isolation made at 10 days interval from 10th day of sowing up to 90th day revealed preferential association of *Fusarium spp.*, *M. Phaseolina* (Tassi) Goid and *Scelrotium rolfsii* sacc.

association of *Fusarium spp.*, *M. Phaseolina* (Tassi) Goid and *Scelrotium rolfsii* sacc.

Katariya *et al.* (2007), studied on genetic variability in *R.bataticola* (*M. phaseolina*) infecting Chickpea isolated using pathogenicity and RAPD Markers and reported that pathogenicity of these isolates varied both in sterilized and unsterilized soils when tested on Chickpea variety C-235.

Thippeswamy *et al.* (2007), studied on pathogenicity and management of brown lesion and leaf spot in Okra caused by *M.phaseolina* and *Alternaria alternata* reported that the pathogenicity test and management was carried out at the department experimental plot. *M.phaseolina*, showed 10-20 per cent spots in one month old seedlings. 1-10 per cent in two month seedling and 10-20 per cent in three months old plant showing brown lesion spots.

2.5.2 Culture media

Mair and Staffeldt (1960), Showed that abundant sclerotial formation of *Rhizoctonia* isolates of Cotton was obtained on PDA.

Bainade *et al.* (1978), reported that the Mung leaf medium was found to be the best for the growth of *Macrophomina phaseolina* over the other media after 4 days, followed by Soybean seed agar media, Richard's agar media, Potato dextrose agar medium, Peptone dextrose, Rose Bengal media were significantly different from each other, while growth on Richard's agar media and potato dextrose agar medium was at par. No growth was recorded on Czapek's agar and Kerr's agar media clearly indicating unsuitability.

Goyal and Mehrotra (1982), reported that trace elements were required for the growth and sclerotia production of *Macrophomina phaseolina* (Tassi) Goid causing dry root rot of Chickpea.

Richard's medium and Groundnut kernel agar and all media were reported well for growth.

Kulkarni *et al.* (1992), reported that *Macrophomina phaseolina* grew best on Cotton root extract, PDA, Richard's agar and Sabouraud's agar.

Suriachandraselvan and Seetharaman (2003), studied effect of culture media on growth and sclerotial production of different isolates of *Macrophomina phaseolina*. Reported that the mycelial growth was highest (89.6mm) in PDA medium followed by Richard's agar (76.7mm) and Oat meal agar (75.1mm) while the lowest growth (57.2mm) was recorded in Czapek's dox agar medium. The sclerotial production as against the highest of 68.5 h in Czapek's dox agar medium.

Salunke *et al.* (2009), experiment was carried out to see the effect of nutritional and physiological factors on growth and sclerotial formation of *R. bataticola* (*Macrophomina phaseolina*) isolates reported that out of six medium PDA was found best medium for growth and sclerotial formation of *R. bataticola* (*Macrophomina phaseolina*) isolates over other media. All isolates showed fast mycelial growth on Malt extract medium and fast mycelial growth on Potato dextrose agar (PDA) followed Czapek's dox agar medium, Richard's agar, Oat meal agar and Peptone agar. No sclerotia production of Soybean isolates on Peptone agar medium.

2.6 Management of Chickpea root rot (*in vitro*)

2.6.1 Integration of biological and chemical seed treatment

Jakhar *et al.* (1997), reported that *T. viride* and *T. harzianum* gives highest germination and Carbendazim recorded lowest germination but at par with *T. viride* *T. harzianum*, and *T. viride* were least effective having disease control of 23.08 and 32.68 per cent respectively, than

Carbendazim which recorded the lowest mortality with control of 65.37 per cent.

Tewari and Mukhopadhyay (2003), studied on management of Chickpea root rot and collar rot by integration of biological and chemical seed treatment at Pantnagar and observed that Chickpea seed treated with *Gliocladium virens* (GC) + Carboxymethyl cellulose (CMC) + Vitavax significantly increased seedling emergence (47.9 per cent); final plant stand (85.8 per cent) and grain yield (79.7 per cent) in sick plot conditions.

Malathi and Doraisamy (2003), reported that inhibitory growth on sclerotial production was significantly increased by the combination of *T.harzianum* and fungicides. Thirum at 10mg/ml showed maximum inhibition of 78 per cent followed by Thirum at 5µg/ml (67per cent and Captan at 10µg/ml (54per cent). The combined effect of antagonist and Captan at 5µ g/ml was not significantly different from *T.harzianum*.

Khan and Gangopadhyay (2008), studied on efficacy of *Pseudomonas fluroescens* in controlling root rot of Chickpea and reported that the root rot incidence was comparatively less in combined treatment of *P. fluroescens* (PFBC-25) with Carbendazim, Carboxin or Chlorothalonil. Further PFBC-26 proved more effective against the pathogen when used in combination with Carbendazim or Chlorothalonil.

Lokesha and Benagi (2007), reported that the four bioagents tested seed treatment + soil application of *T. virens* recorded very less root rot incidence (2.89 per cent) followed by *P. fluroescens* (3.23 per cent) where as 59.13 per cent incidence was recorded in control. However the least disease incidence was recorded in Carbendazim applied as seed treatment + soil drenching (2.41 per cent).

2.6.2 Seed treatment with fungicides

Satischandra *et al.* (1979), studied effect of fungicides on the saprophytic activity of, causing *Macrophomina phaseolina* root rot of beans and reported that Brassicol (PCNB) and Captan were very effective in reducing saprophytic activity of the pathogen where as Carbendazim and Diathane M-45 were not effective.

Taya *et al.* (1990), reported that fungicide reduced dry root rot incidence as compared to control. Carbendazim brought down disease incidence to 28.80 per cent followed by mixture of Carbendazim and Thirum when applied as seed treatment. Out of which eight fungicides applied as soil drench before sowing Carbendazim alone reduced disease incidence from 86.60 to 35.27 per cent followed by mixture of Carbendazim and Thirum (35.50 per cent).

Malathi and Doraisamy (2003), reported that fungicides with increased concentrations significantly inhibited growth and sclerotial production of *Macrophomina phaseolina* by 60 to 100 per cent. Fungicides viz., Thirum (5mg/ml), Captan (5to10µg/ml) were not inhibitory growth while complete inhibition was noticed with Carbendazim (100µg/ml).

Sharma and Gupta (2004), studied efficacy of the fungicide in the control of Chickpea dry root rot caused by (*Macrophomina phaseolina*) and reported that seed treatment with Carbendazim (1 gm/kg seed) showed the minimum incidence of disease (17 to 21 per cent) and increased grain.

Vijay-Mohan *et al.* (2006), studied the fungicidal management of dry root rot disease and yield of Chickpea and reported that Carbendazim (0.2 per cent) and Etaconazole (0.1 per cent) used as seed treatment, soil drenching and combined effect recorded lowest disease incidence of 15.6

and 18.2 per cent and highest grain yield of 192 and 18.9 q/ha, respectively.

Kaushal and Sood (2008), studied management of root rot in Chickpea and reported that seed treatment with Thirum and Bavistin @ 2.0gm/kg seed gave good control of root rot of Chickpea under field conditions.

2.6.3 Seed treatment with bioagents

Parkhia and Vishnav (1986), studied biological control of *Macrophomina phaseolina* and reported that the introduction of *Trichoderma harzianum* reduced the disease to an appreciable level in comparison to check. In seed treatment, 18 per cent plants were infected with *R. bataticola*, in soil drenching 28 per cent and Wheat husk barn culture 14 per cent as compared to 70 per cent in check.

Kehri and Chandra (1991), reported that the antagonist applied as seed coating reduced mortality due to *Macrophomina phaseolina* from 19 to 8 per cent in Mung var T-44 and from 19 to 10 per cent in var Pusa Baisakhi in unsterilized soil under greenhouse conditions. The dry weight of shoots, grains and nodules showed an increase of 31.7, 16.6 and 100.0 per cent respectively in T-44 and 27.0, 32.1 and 133.3 per cent respectively in Pusa Baisakhi.

Selverajan and Jeyaraan (1996), screening eight fungi and bacteria *in vitro* to select suitable antagonists for Chickpea roots rot caused by *Fusarium solani* and *M. phaseolina*. *T. viride*, *T. harzianum*, *T. hamhtum*, *Lectisaria arvalis*, *B. subtilis* and *P. fluorescence* formed inhibition zone against both fungi.

Sankar and Jeyarajan (1996), studied on seed treatment formulations of *Trichoderma* and *Gliocladium* for biological control of *Macrophomina phaseolina* in Sesamum and reported that the no. of colony forming units of *Trichoderma* spp. and *Gliocladium virens* in

fresh product ranged from 224-297x10⁶ gm in different species. The population of antagonist was recorded by 19,21,21,29.9 per cent in *T.viride*, *T.harzianum*, *T. longibrachiatum* and *G. virens* respectively after 75 days of storage.

Singh *et al* .(1998) , studied on application of antagonists in relation to dry root rot and biochemical status in Chickpea plants and reported that all the antagonists used as seed treatment soil inoculation and soil inoculation with seed treatment were significantly superior in control incidence of disease. All antagonists *T. harzianum*, *T.viride*, *P. fluorescence*, *A. flavus* and *B. subtilis* except *A. flavus* were not effective for controlling dry root rot of Chickpea.

Sindhan *et al*. (2002), studied biological control of dry root rot of Chickpea caused by (*Macrophomina phaseolina*) and reported that *Pseudomonas fluroescens* exhibited maximum antagonistic activity causing 65.0 and 90.5 per cent inhibition in mycelial growth and sclerotial production respectively; followed by *Trichoderma viride* which caused 62.3 and 87.0 per cent. mycelial inhibition and sclerotial production, respectively.

Malathi and Doraisamy (2003), reported that *Trichoderma spp.* tested significantly inhibited the pathogen growth and the inhibition ranged from 32 to 67 per cent. The maximum inhibition was shown by *T. harzinum-5* followed by *T.koningii*.

Dhoke and Kurundkar (2004), reported that fungicides with increased concentrations significantly inhibited growth and sclerotial production *Macrophomina phaseolina* on PDA and King's B media. Growth of *Macrophomina phaseolina* was reduced from 55.27 to 67.49 per cent on PDA and from 51.72 to 67.49 per cent on PDA and from 51.72 to 69.43 per cent King's medium.

Sajeena *et al.* (2004), evaluation of biocontrol agents against dry root rot of Blackgram and reported that seed treatment with *Trichoderma viride* and *Pseudomonas fluorescens* resulted in increased seed germination compared to individual treatments and Carbendazim seed treatment.

Gaur *et al.* (2005), reported that optimum storage of talc based formulations form of *Trichoderma* is 90-105 days at 25°C. Then formulations of common organic waste material were tested for growth and multiplication of *T.harzianum* and biocontrol of dry root rot *R. bataticola* (*Macrophomina phaseolina*) of Chickpea. The fecal pellets of goat and sheep, gram shell and farm yard manure to be the best substrate for multiplication (71×10^7 cfu/g) of *T.harzianum* and control dry root rot of Chickpea.

Nair *et al.* (2006), studied on control of root rot of Sesamum by bioagents, and reported that the germination percentage was maximum (92 per cent) in *Trichoderma harzianum* seed treatment, followed by *Trichoderma viride* (89.33 per cent). Similarly per cent root rot incidence was less in treatment with *T. harzianum* (23.18 per cent), followed by *T. viride* (28.35 per cent).

Singh *et al.* (2007), reported that two antagonists *T. harzianum* and *G.virens* as soil application proved to be significantly superior over control in respect to root rot disease control. Highest disease control 66.67 per cent was given by *T. harzinum* followed by *G.virens* (50.00 per cent). *Trihoderma viride* also gave satisfactory results and showed 33.33 per cent disease control.

Khan and Gangopadhy (2008), studied efficacy of *Pseudomonas fluorescens* in controlling root rot of Chickpea and reported that the root rot incidence was significantly reduced due to seed treatment with

PFBC-25 and PFBC-26. Although the disease incidence reduced with an increase in dose from 4 to 8 gm/kg seed.

Choudhary *et al.* (2010), reported that minimum incidence of root rot was observed in *Trichoderma harzianum* treated plants followed by pots treated with *T.viride*. Incidence of root rot was higher in *Bacillus firmus* significantly effective in reducing the dry root rot incidence in Mung bean plants *T. viride* and *T.harzianum* and *A.versicolor* were observed effective in the suppression of dry root rot incidence.

2.6.4 Organic amendments

Ghaffar *et al.* (1969), studied organic amendments on severity of *M.phaseolina* reduced root rot of Cotton was appreciably by amending soil with Alfa-Alfa meal, Barley straw amendment was less effective.

Singh *et al.* (1987), reported that naturally infested soil was amended with residues of mature Wheat, Oat, Chickpea, Peas, and Lentils and inoculated with 10 days old culture of *M. phaseolina*. Seeds JG897 of Chickpea were sown one month after and during the next four month. A gradual decreasing the population of *M. phaseolina* and significantly lowered infection percentage were observed in Wheat and Oat amended with soils.

Hundekar *et al.* (1998), reported that the minimum average infection of 44.45 per cent was recorded with Neem cake amended soils and as such the Neem cake was found to be significantly superior to the other amendments in reducing the disease. The average per cent infection in other amendments were in the Cotton cake, Groundnut cake, Safflower cake, Paddy straw, Wheat straw and FYM respectively, where as 98 per cent infection was recorded with the non-amended soils.

Theradimani and Hepziba (2003), studied on biological control of dry root rot of Sunflower and reported that application of organic amendments such as FYM (12.5 t/ha) and Neem cake (250 kg/ha)

recorded yield 579.50 kg/ha and 645.0 kg/ha respectively. The antagonists combined with organic amendments were more effective than single treatments.

Surichandraselvan *et al.* (2005), reported that application of organic amendments significantly improved seed germination plant growth and seedling vigour. Among these FYM @ 12.5 t/ha was the best followed by Neem cake 250kg/ha. Application of FYM and Neem cake significantly reduced charcoal rot of Sunflower. 13.3 and 15.0 per cent respectively from 36.3 per cent in the control.

Ashraf *et al.* (2007), reported that when soil was amended with different oilcakes and fertilizers, there was significantly increase in fresh and dry weight and it was maximum in diamonium phosphate i.e. 98.6 and 16.3 followed by Mustard cake 90.9 and 18.0 gm, Cotton cake 86.8 and 17.3gm, Neem cake 84.9 and 16.9 gm / urea 87.5 and 14.9 gm and decrease in *R.bataticola* inoculated plant 42.2 and 9.3 gm.

Mallesha *et al.* (2008), reported that lowest disease incidence (18.75 and 31.25 per cent) was recorded in Neem cake at 5 and 2 per cent concentrations followed by *Pongamia* cake (37.5 and 43.75 per cent). Neem cake was found to be significantly superior over all other amendments. Because of release of some inhibitory substances on decomposition affecting population of pathogen.

Sudha and Prabhu (2008), reported that application of organic amendments significantly improved the seed germination, plant growth and seedling vigour. FYM @12.5 tonn/ha was the best treatments followed by Neem cake @250 kg/ha. Significantly reduced Sunflower charcoal rot incidence.

Kaushal and Sood (2008), studied management of root rot in Chickpea and reported that amendment with biogas slurry was found to be very effective in reducing root rot disease incidence it increases

germination 100 per cent and lower disease incidence up to 35.17 per cent.

Jaiman *et al.* (2009), reported that the lowest pre-emergence mortality rot (23 per cent) was recorded with Neem cake followed by Mustard cake. The pooled analysis of two years data revealed that highest yield 76 kg/ha was recorded with Neem cake over the other treatments.

Rajani and Parakhia (2009), reported that the least disease incidence (14%) was in treatment of Neem cake and *T.harzinum* which also gave the maximum Castor seed yield of 2128 kg/ha. Mustard cake and *T.harzianum* were the best treatments similarly all the treatments have increased seed yield over control.

Muhammad *et al.* (2010) reported that applications of Cotton cake and Mustard cake to seed in combination with *Trichoderma spp.* were found to be beneficial in controlling *Macrophomina phaseolina* infection on Sunflower.

2.7 Management of Chickpea root rot (*in vivo*)

2.7.1 Chemical Control

Sinha and khare (1977), studied on chemical control of *Macrophomina phaseolina* and *Fusarium equiseti* associated with Cowpea seeds and reported that seed treatments trail against *Macrophomina phaseolina* in pots reveal that pre-emergence losses were maximum (38 per cent) with Thiabenadazol and minimum (9 per cent) with Thirum as against 41 per cent loss in control. The mortality was 2, 3, 5 and 6 per cent in the treatment of Captan and Thirum, Bivistin, Ceresin dry, Difolatan and Thibendazole, Brassicol and Phatan and control respectively. Total mortality being maximum (41 per cent) with Thibendazole and minimum (11 per cent) with Thirum against 47 per cent control.

Rathore and Rathore (1999), studied on effect of seed dresser on *Macrophomina phaseolina* root rot incidence of Moth bean and reported that, the highest per cent plant emergence (95.0 and 90.4 per cent) was recorded when Bavistin 50 WP (2 gm/kg) pre treated seeds were sown, followed by closely the seed treatment with Captan 75 SD (3 gm/kg), Topsin M-70 (2 gm/kg) and Subeej 25 SD (2 gm/kg). Lowest post emergence plant mortality (6.1 and 19.5 per cent) was recorded in plot where as seeds were sown after treatment with Bivistin 50 WP (2 gm/kg) followed by Captan 75 SD, Topsin M70 and Subeej 25 SD.

Prajapati *et al.* (2004), studied incidence of Chickpea dry root rot in Uttar Pradesh and efficacy of seed dressing fungicides on seed germination and seedling infection and reported that application of Carbendazim in combination with Thiram resulted in the highest seed germination percentage (71.4%) and lowest root rot incidence (0.59 per cent) which was at par with Carbendazim spraying.

Bagri *et al.* (2004), reported that Thiram gave the highest seed germination (92 per cent) and minimum pre and post emergence mortality (6 and 3 per cent respectively) with less number of seedling showing symptoms of *Macrophomina phaseolina* was also effectively controlled by Thiram which also showed the highest seed germination 89 per cent and minimum post emergence mortality (7and 3% respectively).

Jaiman and Jain (2004), studied on *Macrophomina phaseolina* in Cluster bean seeds and its control reported that minimum per-emergence rot, post-emergence rot, disease incidence and maximum grain yield observed in Carbendazim, Raxil and Indofil M-45 were found inferior almost in all treatments.

2.7.2 Biological Control

Lynch (1989), outlined the mechanism by which fungi can antagonize plant pathogens in *vitro* and *in vivo* and brief analysis

opportunities for gaining a better understanding of fungal system was presented with emphasis *Trichoderma spp.* as antagonists.

Saxena and Saxena (1995), studied on evaluation of antagonist against dry root rot pathogen of Chickpea reported that seed treatment and soil application of *T.viride* was significantly effective in controlling root rot up to 96.0 per cent and gave 38.37 gm/plot followed by its soil application alone. Seed coating and soil application of *T. harzianum* was effective and reduced root rot up to 84 per cent. Seed coating with *Bacillus subtilis* and soil application of *G.mosseae* individually checked the disease incidence 52.0 and 44.0 per cent respectively.

Upadhyay *et al.*(2000), evaluated several antagonism viz., *Trichoderma spp.* *Gleocladium virens*, *Bacillus spp.*, *Pseudomonas fleuroscence* *Aspergillus spp.* and *Penicillium spp.* *in vitro* and *in vivo* against soil borne pathogens *Fusarium*, *Rhizoctonia*, and *Macrophomina* in different pulses like Chickpea, Lentil, Pea, Pigeonpea and Mung bean.

Indra and Subbiah (2003), reported that seed treatment with *Trichoderma spp.* alone or in combination with biofertilizers significantly reduced the root rot incidence by 50 per cent both under glasshouse and field condition when compared to control. At 75 days least disease incidence *T. viride* + *Rhizobium* (40.93per cent) *T.harzinum* (biomass) + *Rhiaobium* (42.16percent) *T.harzianum* (gypsum) + *Rhizobium* (43.5 per cent) and *T.harzianum* (talc) + *Rhizobium* (43.83 per cent). Maximum incidence of 93.13 per cent was recorded in untreated control.

Theradimani and Hepziba (2003), studied on biological control of dry root rot of Sunflower reported that *T. viride* and *T. harzianum* were tested against root rot of Sunflower caused by *Macrophomina phaseolina* for two consecutive season.

Surichandraselvan *et al.* (2004), studied on seed treatment with *Trichoderma spp.* for the control of Sunflower charcoal rot caused by *Macrophomina phaseolina* and reported that seed treatment with talc based formulation (4 gm/kg) of *T. viride* or *T. harzianum* significantly reduced the disease incidence (22-27 per cent) compared to untreated control 67 per cent.

Gaur *et al.* (2005), studied on *Trichoderma* for *Rhizoctonia* root rot control in Chickpea can be effectively and economically managed either by soil application of 10-15 days pre-incubated *T.harzianum* (TG-1) @ 10 kg/ha in 200 kg of FYM or by the practice of seed treatment with talc-based formulations of *T.harzianum* containing 2×10^6 CFU/g @ 10 g/kg seed in combination with soil application of $ZnSO_4$ @ 25 kg/ha under rainfed situation seed treatment with talc based formulations of *T. harzianum* @ 10 kg/ha seed proved economically better.

Sundravadana and Alice (2006), reported that $ZnSO_4$, *T.viride* and *Pseudomonas fluorescens* induced Peroxidase (PO) and Polyphenol Oxidase (PPO) activity in Black gram root challenged with root rot pathogen.

Bhattacharyya and Ghosh (2006), studied on *in vivo* effect of *P.fluorescent* on *Macrophomina phaseolina* causing root rot in Brinjal reported that some of the *fluorescent* bacterial isolates have antifungal properties against *Macrophomina phaseolina*. The isolates KBGA *in vivo* condition successfully controlled root rot of Brinjal.

Vanitha and Alice (2008), reported that application of vermiculite along with seed treatment of *T. viride* (1335.53 kg/ha) with cost benefit ratio 1:2.5. Treatment comprising seed with dry Neem leaf powder @ 100 gm/kg seed + *T.viride* @ 4 gm/kg of seed also showed better performance by recording 13.5% root rot infection with 1206 kg/ha seed. Grain yield and CB ratio of 1:2 application of proposed leaf dry powder

@ 100 g/kg + *T.viride* 4 gm/kg seed recorded 15.2 per cent root rot infection and registered the yield 1138 kg/ha with CB ratio of 1:2.

Jaiman *et al.* (2009), studied on field evaluation of fungicides, bioagents and soil amendments against root rot caused by *Macrophomina phaseolina* in Clusterbean reported that a minimum disease incidence (25per cent) and pre-emergence rot were observed in *T.viride* which was closely followed by *T. harzianum* treatment. A maximum yield of Clusterbean was also found superior with *T.viride* (763 kg/ha).

2.8 Screening of Chickpea varieties, germplasm lines and genotypes

Agarwal and Sarbhoy (1976), studied on evaluation of Soybean germplasm for resistance against *Macrophomina phaseolina*, reported that sixty-four lines were found to moderately resistant and four lines viz., EC-1077, EC-25166, EC-39069 and EC-39797 resistant under artificially inoculated field as well as glass house condition.

Nene *et al.* (1989), reported that at each of the 24 locations, few to several lines were found to be resistant to wilt and root rots in Chickpea. In addition, the genotypes ICC-9023, ICC-9032, ICC-10803, ICC-11550, and ICC-11551, were resistant to wilt and root rot at 11 to 12 locations of 24 locations where these were tested.

Singh *et al.* (1989), studied on evaluation of Sesamum germplasm for resistant against *Macrophomina* root rot reported that out of 312 cultivar lines were tested in which 30 lines were found resistant; 158 moderately resistant; 124 showed susceptible reactions. The resistant lines were AK-77, ES-4, IS-523, IS-175, KIS-430, N66-3169, SI-1740, SI-1153, SI-1875.

Reddy *et al.* (1990), reported additional sources to wilt and root rot in Chickpea from ICRISAT, Hyderabad out of which 1883 accessions were evaluated in which 177 accessions showed less than 20 per cent

mortality in the first year of screening and of these 40 accessions showed less than 10 per cent mortality in second year of screening.

Reddy *et al.* (1991), studied on evaluation of wild *Cicer* accession for resistance to wilt and root rot nursery at ICRISAT centre between 1986-1988. Annigiri and JG-62 cultivars were used for comparison. Some wild species showed equal or less mortality than Annigiri and remaining showed higher mortality.

Baker and Ahmed (1991), studied on additional source of resistance to wilt and root rot of Chickpea in Bangladesh and reported that 12 entries were selected from 18 entries. These 12 entries were again screened in a sick plot during 1990-1991 and finally selected as resistant to wilt and root rot. Mortality of susceptible entry ICC4951 was 100 per cent during the screening.

Lodha and Solanki (1993), studied on inheritance of resistance to dry root rot in Cluster bean and reported that the susceptibility in the genotypes varied from 22.3 to 46.6 per cent. However, least (26.1%) pre-emergence mortality was recorded in the HFG-156 followed by IC-922913 (33.9%) while Kutch-8 recorded lowest (22.3%), Post emergence mortality followed by RGC-471 and highly susceptible varieties HG-75 (40.5%) and AG -111 (46.6%) were identified for their use in resistant breeding programme.

Zote *et al.* (1993), screened Chickpea germplasm lines against wilt and dry root rot pathogen and reported BCP-28, BCP-29 and BCP-36 lines were resistant to *Fusarium* wilt and lines BCP-15, BCP-23, BCP-29 and BCP-30 were resistance to both wilt and dry root rot.

Diodrte *et al.* (1995), studied on charcoal rot *Macrophomina phaseolina* resistance and the effects of water stress on disease development in Sorghum reported that two Sorghum genotypes B35-6 and SC-265-14E were found to be consistently more resistant to

Macrophomina phaseolina an assessment that was facilitated by the use of appropriate soil moisture condition at the time of greatest plant susceptibility to charcoal rot.

Prajapati *et al.* (2003), forty five Chickpea cultivars were evaluated for resistance to dry root rot (*Macrophomina phaseolina*). Among these following cultivars were tested as resistant, H-355, JG-315, Pusa-212, ICC-32, BG-244, Avrodhi, and ILC-702. Five cultivars moderately resistant (GCP-9504, Phule G numbers 96022, 96020 and GL 91059) and 9 were tolerant (JG-22, JG-23, Phule G-96022, 95123, 93208, 92028, BCP-91 and BCP-28).

Pande *et al.* (2004), studied on evaluation of Chickpea lines for resistance to dry root rot caused by *R. bataticola* and reported that, one germplasm accession (ICC-14995), a cultivar (ICCV-2) and an advanced breeding lines (ICCX830203-BH-BH-11H) were resistant to dry root rot of the remaining 22 lines were moderately resistant, 19 susceptible and 3 highly susceptible.

Sajeena *et al.* (2004), studied on screening of Blackgram varieties against root rot caused by *Macrophomina phaseolina* reported that among the seventeen genotypes of Black gram tested BG ADT 5 and BGKI recorded the maximum germination and yield and root rot disease incidence was at a minimum. IC14520 and IC14691 registered the minimum germination and yield and showed maximum disease incidence. None of the genotypes were found to be completely resistant to root rot disease.

Dinkaran *et al.* (2005), studied on screening of Sesame genotypes against root rot pathogen *Macrophomina phaseolina* reported that among the 1300 Sesame genotypes screened only three accessions viz., RM-7, ORM-14 and ORM-17, showed resistant reaction.

Desai (2005), studied on field evaluation of advanced Sorghum hybrid against charcoal rot disease caused by *Macrophomina phaseolina* conclude that CSH-13R, SPH-641, SPH-634, and SPH-677 were Sorghum hybrid with charcoal rot tolerance. Host plant resistance of Sorghum against charcoal rot disease viz., crosses (CM-35-1x CSV-8), (CM-35-1x E-36-1) and (CSV-8Rx E-36-1); B-35-6, SC-265-11E and IS-443 and IS-3547 SPH-296 and SPH -509(5).

Mohmad *et al.* (2005), reported that 56 Chickpea varieties tested and none of the variety was found immune to *M. phaseolina* on the basis of suppression in dry shoot weight and root rot index. One variety (BGD-98) was found as resistant, three moderately resistant (BG-1108, BGD-117 and RSG-143) and 6 varieties (BG-390, BG-1095, Biogreen, GPF-2, ICCV-92337 and SBD-77) were found tolerant to *M. phaseolina*. The varieties viz., BG-217, BG-267, BG-1006, BG-1032, BG-1072, BG-1082, BG-1086, BG-1087, BG-1107, BG-10863, BGD-7, BGD-112, CSG-9505, EC-442360, EC-4422507, ICC-88503, ICCV-5, KPG-59, L-550, Pusa-212, Pusa-362, Pusa-391, Pusa-1053 and RYT-411 were found susceptible and the remaining varieties viz., BG-276, BG-376, BG-1044, BG-1063, BG-1079, BG-1091, BG-1100, BG-1106, BG-112k1, BGD-1104, C-235, CIYTSL-2, EC-44285, ICC-8803, ICCV-2, Pusa-256, Pusa-372, Pusa-1003, Pusa-1062, SAKI-9303 and Vijay were rated as highly susceptible to *M. phaseolina*.

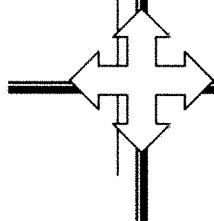
Jaiman *et al.* (2006), studied on field evaluation of Cluster bean genotypes against root rot pathogen *Macrophomina phaseolina* reported that genotypes showed resistance reaction (1-10 % mortality). They included CAZG-99-24, CAZG-986, CAZG-9819, HGS-832, ROC-986, RGC-1003, ROC-8-14, ROC-76, RGC-1020, and RGM-105. Other genotypes showed moderately resistant reactions (11 to 20 % mortality). The genotypes were HGS -827, ROC-102, RGC-1002, RGC-8-35/P,

RGC-1017, and ROM-112, Cluster bean genotypes CAZG-9-1, RGC-10-18 ROC-936, HG-346, AVG - 9102 were susceptible to root rot disease.

Vijay-Mohan *et al.* (2006), reported that Chickpea lines/cultivars G-543, BG-360, GNG-1365 were resistant, Pusa-372, RSG-888 and RSG-931 showed resistant reaction against dry root rot disease under natural conditions. Under artificial conditions, cultivars G-543, BG-360 and GNG-1292, RSG-888, RSG-895, H-208 showed susceptible reaction against disease.

Katariya and Gaur (2007), studied on reaction of Chickpea varieties for resistance to dry root rot caused by *R. bataticola* (*Macrophomina phaseolina*) reported that out of twenty three varieties screened against dry root rot two varieties viz., BG-203 and BG-209 which recorded incidence below 10 percent were termed as resistant to dry root rot . BG-377, BG-416, BG-1053, GMS-469, GMS-1292, GNG-469, RSG-44 and RSG-888 having incidence below 11 -30 percent were categorized moderately resistant. Twelve varieties viz., BG-256, BG-1103, BG-ICC-4951, RSG-897, RSG-963 and RSG-974 had 31 to 70% disease incidence termed as moderately susceptible. Variety C-235 had incidence of disease up to 100 percent was termed as highly susceptible.

*MATERIAL
AND
METHODS*



Chapter III

MATERIALS AND METHOD

During the present investigations on dry root rot of Chickpea (*Cicer arietinum* L.), caused by *Macrophomina phaseolina*. A number of experiments were planned and conducted (*in vitro* and *in vivo*) during Rabi, 2010 at the Department of Plant pathology, College of Agriculture, Latur to fulfill the objectives of the present study. Details of the materials used and methods followed are presented in the following paragraphs.

3.1 Experimental Materials

The following materials were used during this experiment.

3.1.1 Disease sample

Plant showing typical symptoms of Chickpea dry root rot were collected from farmer's field belonging to eight tahsil of Latur district and preserved.

3.1.2 Culture Media

Potato dextrose agar (PDA), the common laboratory medium was used for isolation of the pathogen associated with dry root rot disease of Chickpea and also several other culture media viz., Sand : Maize (1:1) meal medium, were used for mass multiplication of pathogen and Oat meal agar, Richard's medium, Asthana and Hawaker's media, Czapek's dox agar medium, Host extract agar media and Potato dextrose agar (PDA), were used to study their effects on mycelial growth and sclerotial production of *M. phaseolina*.

3.1.3 Seed

Seed of Chickpea varieties, cultivars and germplasm lines collected from ARS (Pulses), Badnapur required for various studies was used in present investigations.

3.1.4 Chemicals

Standard chemicals, reagents, fungicides, etc. and chemicals required for the preparation of different culture media were obtained from Department of Plant Pathology, College of agriculture Latur.

3.1.5 Biocontrol Agents

Biocontrol agents like *T. harzianum*, *T. viride*, *T. hamatum*, *Pseudomonas fluorescence* were obtained from Department of Plant Pathology College of Agriculture Latur.

3.1.6 Fungicides

The following fungicides were used *in vivo* and *in vitro* (potculture) against *M. phaseolina*.

1. Thiram 75 % WP
 - a. Trade name - Thiram.
 - b. Active ingredient - 75 % Wettable powder.
 - c. Concentration used - 0.3 % (3 gm/kg seed).
 - d. Manufacture - Coromandel fertilizers Ltd.
2. Carbendazim 50 % WP
 - a. Trade Name - Bavistin.
 - b. Active ingredient – 50 % Wettable powder.
 - c. Concentration used - 2 gm/kg seed.
 - d. Manufacture (S) - BASF, India Ltd., Mumbai.
3. Mancozeb 75 % WP
 - a. Trade Name - Diathane M-45.
 - b. Active Ingredient – 75 % Wettable Powder.
 - c. Concentration Used - 1.5 gm/kg seed.
 - d. Manufacturer (S) - Dow agro sciences, P.Ltd., Mumbai.
4. Captan 50 % WP
 - a. Trade Name - Captaf.

c. Concentrations Used - 3 gm/kg seed.

d. Manufacturer (S) - Rallies India Ltd., Mumbai. T 6244

3.1.7 Organic Soil Amendment

Various oil seed cakes were obtained from the Latur Local market and their details are as follows:

- Groundnut Cake (*Arachis hypogea*) : 10 g/kg soil.
- Neem Seed Cake (*Azadirachta indica*) : 10 g/kg soil.
- Cotton Seed Cake (*Gossypium spp.*) : 10 g/kg soil.



3.1.8 Glassware's

Different type of Corning and Borosilis brand glassware's were used in the present investigations. The common glassware's used were Petri plates, test-tubes, conical flasks, pipettes, beakers, glass rod, spirit lamp and measuring cylinders.

3.1.9 Equipments

The laboratory equipments used such as Autoclave, Laminar Air Flow Cabinet, BOD incubator, Refrigerator, Shaker, Research Microscope, Isolation Chamber and Analytical Scientific Balance.

3.1.10 Miscellaneous Materials

Inoculating needle, forceps, Blotter paper, Spirit lamp, Mercuric chloride ($HgCl_2$), distilled water, earthen pots (25), water cans, Cotton, Paper bags, Polythene bags, labels, scales, soil, sand, Formalin, Maiz seeds were used during the course of present investigations.

3.2 Methodology

3.2.1 Survey

A field survey of Chickpea dry root rot disease was conducted in eight tahsils of Latur district during the month of December and February, 2009-10 to record occurrence and distribution of Chickpea dry root rot disease. On an average 10 farmer's field of Chickpea in each tahsils were visited and the per cent dry root rot disease incidence was

recorded by counting total Chickpea plant on $1 \times 1 \text{ m}^2$ area and total dry root rot infected plants in that area. Chickpea plants showing typical symptoms were collected in separate paper bags and brought to the laboratory for investigations.

3.2.2 Isolation, Pathogenicity, Reisolation and Symptomatology.

3.2.2.1. Isolation

Diseased Chickpea plants showing seedling mortality and typical dry root rot symptoms were collected during the survey of Latur district and brought to the laboratory for further studies. The roots of diseased plants were rinsed in running tap water to wash adhered soil particles. Root showing dry root rot infections were cut into small bits with sharp sterile blade and surface sterilized with (0.1 per cent) aqueous solution of Mercuric chloride (HgCl_2), then washed by giving three serial changes with sterilized water to remove traces of the Mercuric chloride (HgCl_2) and blot dried. These bits were then placed on the autoclaved, solidified and cooled Potato Dextrose Agar (PDA) medium in petriplates under aseptic conditions of Laminar air flow cabinet. These plates were then incubated in BOD incubator at $27 \pm 1^\circ \text{C}$ temperature for a week. The well-developed mycelial growth free of contaminant obtained was then transferred to agar slants by single hyphal tip method. Through frequent sub culturing, the test fungus was purified and maintained on agar slant for further studies.

3.2.2.2. Pathogenicity

Pathogenicity of the test fungus was confirmed by soil inoculation method in earthen pots.

Pure culture of *Macrophomina phaseolina* was multiplied on sand : maize meal medium (sieved fine river sand 100g + maize meal 80 g + distilled water 150 ml) for 10 days and uniformly mixed (@ 100 g/kg soil) with sterilized potting mixture of soil : sand: FYM (2:1:1).

Twenty earthen pots (25 cm diameter) were disinfected with 5% Copper sulphate solution. Pots were filled with potting mixture and inoculated with mass culture of *Macrophomina phaseolina*. One pot filled with sterilized potting mixture and without culture of *Macrophomina phaseolina* was maintained as uninoculated control. These pots were incubated for 15 days at room temperature, frequently stirred, watered regularly and allowed the fungus to colonize better in the pot soil. Then surface sterilized seeds of Chickpea Cv. JG-62 were sown (@ 10 seeds/pot) in all these pots and kept at room temperature and watered regularly. Observations were recorded on number of dry root rot infected Chickpea plants.

3.2.2.3. Reisolation

The pathogen was reisolated on basal medium PDA from the root region of the artificially inoculated Chickpea seedling showing seedling mortality and typical symptoms of root rot. Growth of the test fungus, obtained was transferred on Potato dextrose agar slants and compared with original pure culture of the test fungus obtained from naturally root rot diseased plants.

3.2.2.4. Identification of the pathogen

Based on morphological and cultural characteristics, microscopic observations and pathogenicity test, the test pathogen under investigation was identified and confirmed as *Macrophomina phaseolina* (Tassi) Goid from Agharkar Research Institute, Pune.

3.2.2.5. Cultural studies

The cultural characteristics viz., colony characters, morphology, mycelial growth and sclerotia production of *Macrophomina phaseolina* were studied on different culture media. A total of six culture media viz., Oat meal agar, Richard's agar, Asthana Hawkers media, Czapek's dox agar, Host leaf extracts, Potato dextrose agar, were used. Autoclaved and

Oat meal agar, Richard's agar, Asthana Hawkers media, Czapek's dox agar, Host leaf extracts, Potato dextrose agar, were used. Autoclaved and cooled (40°C) media were poured into sterilized glass petriplates (90 mm diameter) and allowed to solidify. Upon solidification of the media, plates were inoculated aseptically with 5 mm culture disc obtained from 5 - 6 days old culture of *Macrophomina phaseolina*. Each treatment was replicated thrice and plates were incubated at $30 \pm 2^{\circ}\text{C}$.

Details of experiments were as follows.

Design	:	CRD
Replications	:	Three
Treatments	:	six

Treatment Details.

1. Oat meal agar.
2. Richards's agar.
3. Asthana Hawakers media.
4. Czapek's dox agar.
5. Host leaf extract agar.
6. Potato dextrose agar.

3.2.2.5.1. Observations

Observations on radial mycelial growth, sclerotia production and cultural characteristic were recorded.

3.3. Evaluation of fungicides, bioagents and organic amendments against *Macrophomina phaseolina* (Pot culture).

Pot Culture

a) Layout

Design	:	RBD
Variety	:	JG-62
Replications	:	Three
Treatments	:	Ten

Treatments details (Seed treatments and soil application)

Treatments	Details	Dosage g/kg
T1	Thiram (75 % WP)	(@ 3 g/kg seed)
T2	Carbendazim (50 % WP)	(@ 2 g/kg seed)
T3	Captan (50 % WP)	(@ 3 g/kg seed)
T4	<i>Trichoderma harzianum</i>	(@ 10 g/kg seed)
T5	<i>Trichoderma viride</i>	(@ 10 g/kg seed)
T6	<i>Trichoderma hamatum</i>	(@10 g/kg seed)
T7	Neem cake	(@500 kg/ha soil)
T8	Cotton cake	(@500 kg/ha soil)
T9	Groundnut cake	(@500 kg/ha soil)
T10	Control(Untreated)	

In fungicidal trial, the seed of the variety JG-62 were treated with Thiram (3 gm/kg seed), Carbendazim (2 gm/kg seed), Captan (3gm/kg seed) individually (dry seed treatment) and sown (@ 10 seeds/pot) in earthen pots disinfected with 5% Copper sulphate and containing *Macrophomina phaseolina* sick soil. The pots were watered periodically.

The seeds of the variety JG-62 were treated with Bioagent *Trichoderma harzianum* (10 gm/kg seed), *Trichoderma viride* (10 gm/kg seed), *T.hamatum* (10 gm/kg seed), against *Macrophomina phaseolina*. Prior to use, earthen pots were disinfected with 5% Copper sulphate. The treated seed of variety JG-62 were sown (@10 seeds/pot) in these pots containing *Macrophomina phaseolina* sick soil.

Sterilized earthen pots (25 cm diameter) were filled with autoclaved potting mixture of soil : sand (1:1) and as per treatment details, potting mixture was treated with powdered seed cakes (each @ 10

gm/kg soil) before seven days of inoculation with test pathogen and watered periodically. Then these pots were inoculated (@ 80 gm/kg soil) with mass culture of test fungus (*Macrophomina phaseolina*) multiplied on sand: Maize medium (1:1), watered periodically and incubated at room temperature for one week. The seeds of Chickpea cultivar JG-62 were sown (@ 10 seeds/pot), in the pots and watered periodically. Ten seeds of variety JG-62 were sown in earthen pots containing untreated potting mixture but inoculated with test fungus were maintained as control. Observation on per cent seed germination, no. of infected plant were recorded at 15 days intervals from germination up to mortality

3.3.1 Observations

Observations on germination count and number of plants rotted per replications were recorded at 30, 45, 60 and 75 days after sowing. The per cent dry root rot incidence was calculated on the basis of initial plant count and total number of rotted plants.

3.4 *In vivo* evaluation of fungicides, bioagents condition

3.4.1 Field evaluation

The field experiment was conducted (fig.1) during *rabi* 2010 on research farm of the Department of Plant Pathology, College of Agriculture, Latur. Fungicides *viz.*, Thiram, Carbendazim alone Carbendazim 50 % WP + Thiram 75% WP, and Carbendazim 50 % + Mancozeb 50 % WP in combination and bioagents *viz.*, *T.harzianum*, *T. viride*, *Pseudomonas fluorescence* alone and *T. harzianum* + *T. viride* in combination, were evaluated against *Macrophomina phaseolina*, all these treatments applied before sowing. One plot per replication was maintained without application of any treatment as untreated control. The details of the field layout and treatments are as follows.

a) Layout

Design	:	RBD
Variety	:	JG-62
Replications	:	Three
Treatments	:	Nine
Plot size	:	3 x 2 m
Spacing	:	30 x 10cm

Treatments details (Seed treatments)

Treatments	Details	Dosage g/kg
T1	Thiram (75 % WP)	(@ 3 g/kg seed)
T2	Carbendazim(50 %WP)	(@ 2 g/kg seed)
T3	Carbendazim (50%WP)+ Thiram(75 % WP)	(@1g+1.5 g/kg seed)
T4	Carbendazim(50%WP) + Mancozeb(75 % WP)	(@ 1 g + 1.5 g/kg seed)
T5	<i>T. harzianum</i>	(@ 10 g/kg seed)
T6	<i>T. viride</i>	(@ 10 g/kg seed)
T7	<i>Pseudomonas fluroescens</i>	(@ 10 g/kg seed)
T8	<i>T.harzianum</i> + <i>T. viride</i>	(@ 5 g + 5 g/kg seed)
T9	Control (untreated)	

3.4.2. Observations

Observations on number of plants rotted per replication were recorded at 30, 45, 60 and 75 days after sowing. The per cent dry root rot incidence was calculated on the basis of initial plant count and total number of rotted plants. Plot wise seed yield and test weight was also recorded.

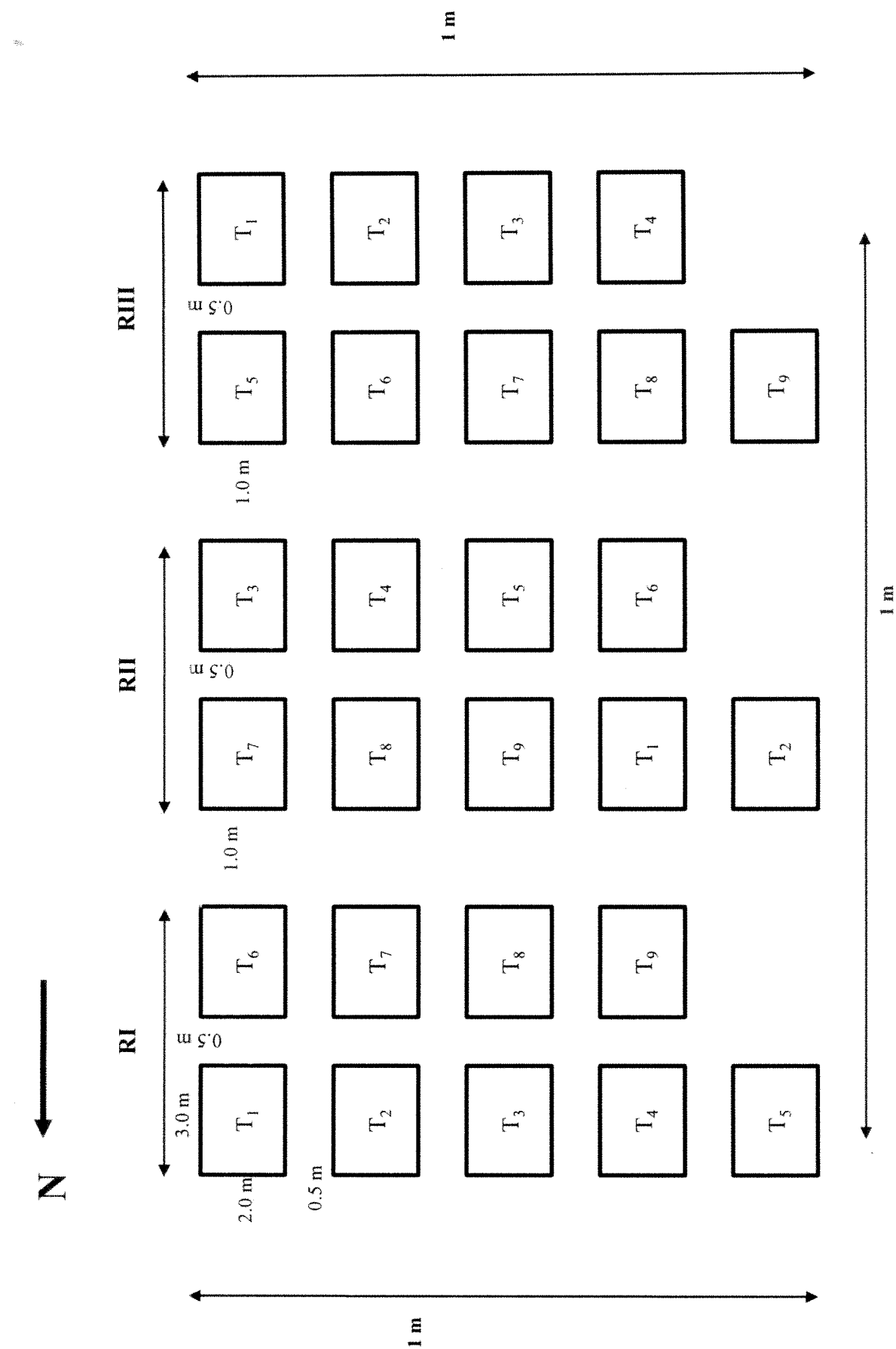


Fig.1 Plan of Layout of field experiment evaluation of fungicides and bioagents against *M. phaseolina*

3.5 Varietal / Germplasm screening

3.5.1 Pot screening

Promising genotype and breeding material were tested in pot containing sick soil (*Macrophomina phaseolina*) to confirm the reaction of Chickpea genotype against the pathogen.

To identify the sources against dry root rot disease resistance in Chickpea varieties, cultivars and germplasm lines, screening (pot culture) was undertaken at the Department of Plant Pathology, during *Rabi*, 2010.

Disinfected earthen pots were filled with sterilized soil and inoculated with sand: maize base culture of *Macrophomina phaseolina* incubated and watered regularly to make sick soil.

3.5.2 Observations

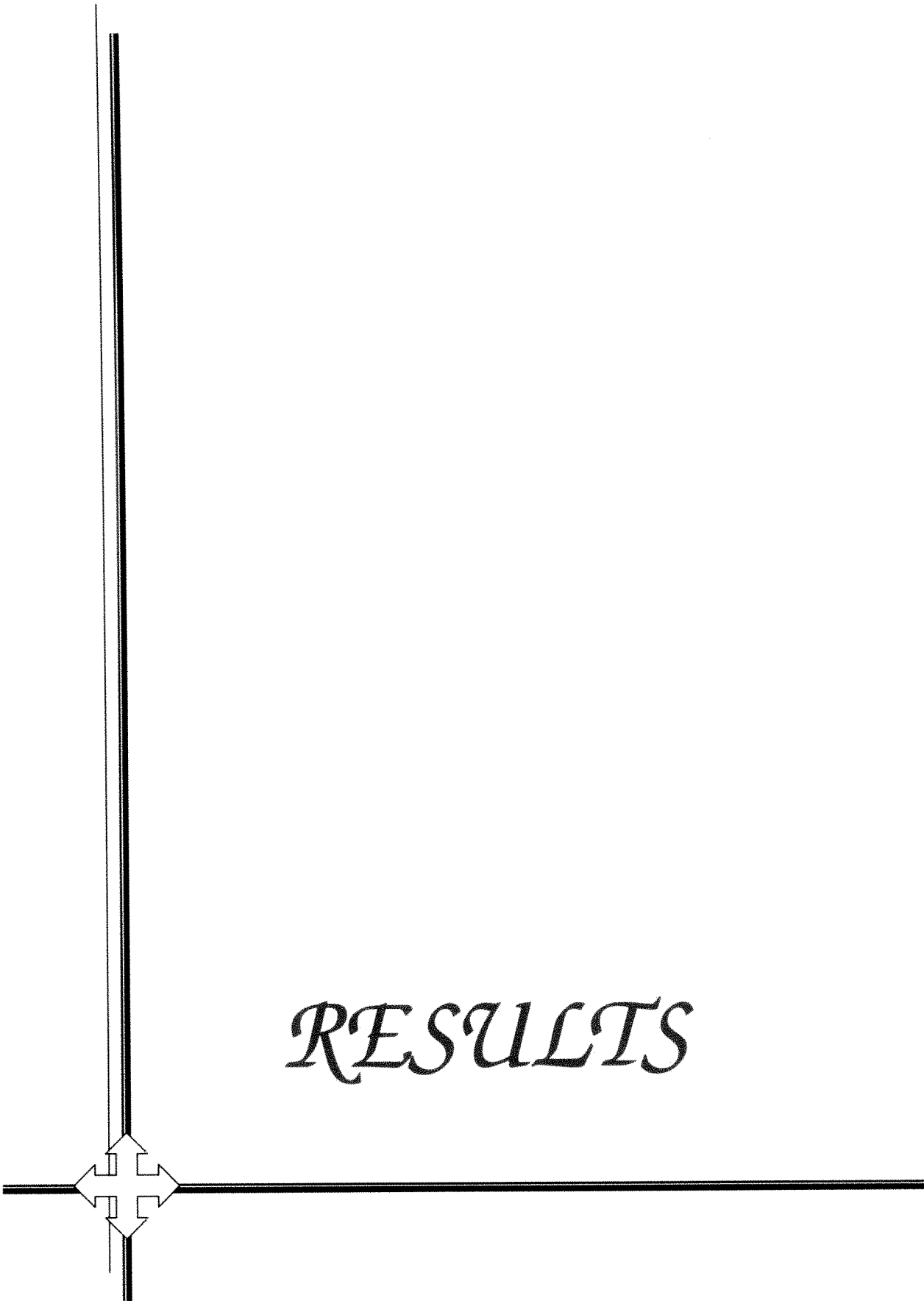
Observations on number of plants rotted from each genotype were recorded at 30, 45, 60 and 75 days after sowing. The per cent root rot incidence was calculated on the basis of initial plant count and total number of rotted plants in each genotype and genotypes were graded as by applying disease rating scale.

Disease reaction	Per cent disease incidence
1. Resistant (R)	: 0.00-10.0
2. Moderately resistant (MR)	: 10.1-20.00
3. Susceptible (S)	: 20.1- 50.00
4. Highly susceptible (HS)	: 51 or more

3.6 Statistical analysis

The experimental data was statistically analyzed on SAMSUNG model at Computer centre, College of Agriculture, Latur. Observation of per cent disease incidence was converted in value of arcsine transformation for statistical analysis.

RESULTS



Chapter IV

RESULTS

Present studies on *Macrophomina phaseolina* (Tassi) Goid infesting Chickpea (*Cicer arietinum* L.) were undertaken during *Rabi* 2010, on the aspect viz., Survey on dry root rot, isolation, pathogenicity and cultural characteristics of pathogen, *in vitro* evaluation of fungicides, bioagents and organic amendments (pot culture), *in vivo* evaluation of fungicides and bioagents and germplasm screening. The results obtained on all the aspects are being presented in the following paragraphs.

4.1 Survey

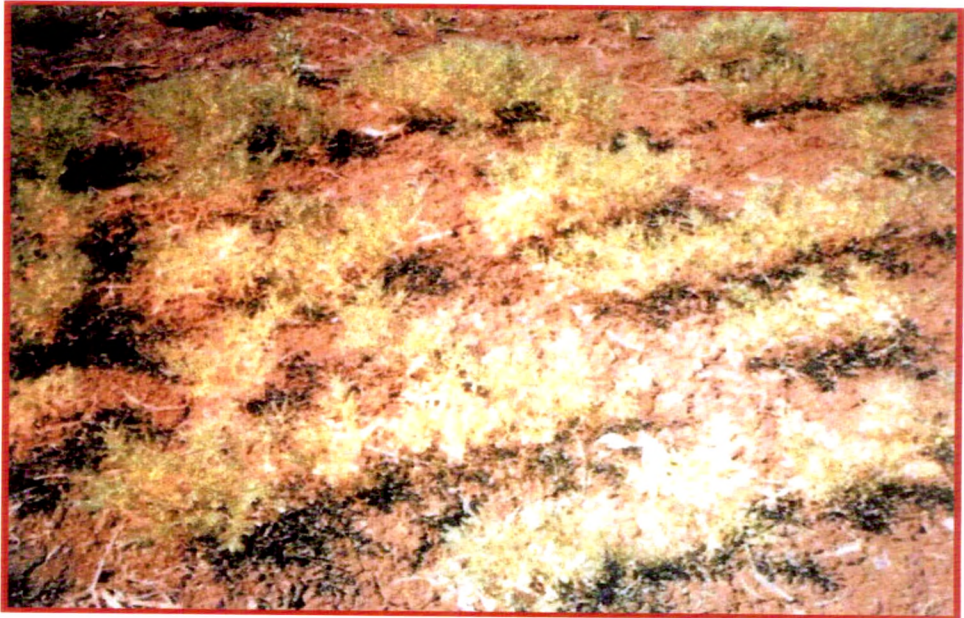
A roving survey for recording Chickpea dry root rot disease incidence was undertaken in Latur district of Marathwada region. A total of eighty Chickpea fields of the farmers in the eight tahsils (Latur, Jalkot, Chakur, Ahmedpur, Ausa, Nilanga, Renapur and Udgir,) of the Latur district were traversed and dry root rot disease incidence were recorded and data is presented in Table-1. Heavy and moderate disease incidence in Ausa and Latur district showing in (Plate-I) and (depicted in fig.2).

Table 1: Survey of Chickpea dry root rot disease from farmer's field.

Sr. No.	Name of Tahsil	No. of location traversed	Locations	Av.% of dry root rot disease incidence in each tahsils
1.	Latur	12	Pakharsangvi, Bamni, Kolpa, Pedgaon, Bhatangli, Gharni, Takali, Malwadi, Bhatkheda, Shirala, Gangapur, Sirsi.	14.21

2.	Jalkot	10	Hokharna, Umberda, Keksindgi, Seheldara, Dhamangaon, Dongaon, Mangrul, Wanjarwada, Jirga, Nalgir.	9.60
3.	Chakur	7	Hadoli, Zari, Nalegaon, Achola, Ashtha, Belgaon, Lari.	11.50
4.	Ahmedpuar	8	Kingoan, Nandura, Ujna, Vilegoan, Hiwarebazar, Tajband, Shirur, Godhalo	18.92
5.	Ausa	10	Bhusni, Bhada, Aalamla, Tungi, Devtala, Killari, Nagarsopa, Dapegaon, Jawalgwadi, Haregoan.	21.60
6.	Nilanga	9	Hisamnagar, Malegoan, Hadga, Nithur, Hangarga, Tagarkheda Ambulga, Kedarpur, Sonkhed.	9.15
7.	Renapur	15	Sangavi, Andalgaon, Bitergaon, Pangaon, Karepur, Kharola, Kamkheda, Harwadi, Faradpur, Sumthana, Mohgaon, Darji borgaon, Bhokarmaba, Pohergaon, Ghanshargaon.	13.00
8.	Udgir	9	Astamode, Dongarshelki, Haknakwadi, Karadkhel, Hali Handarguli, Nagalgaon, Togri, Wadhona, Telegaon.	12.20
	Total	80	Mean disease incidence	13.85

PLATE - I



AUSA



LATUR

PLATE-I Heavy and moderate incidence of dry root rot on farmers field in Latur district.

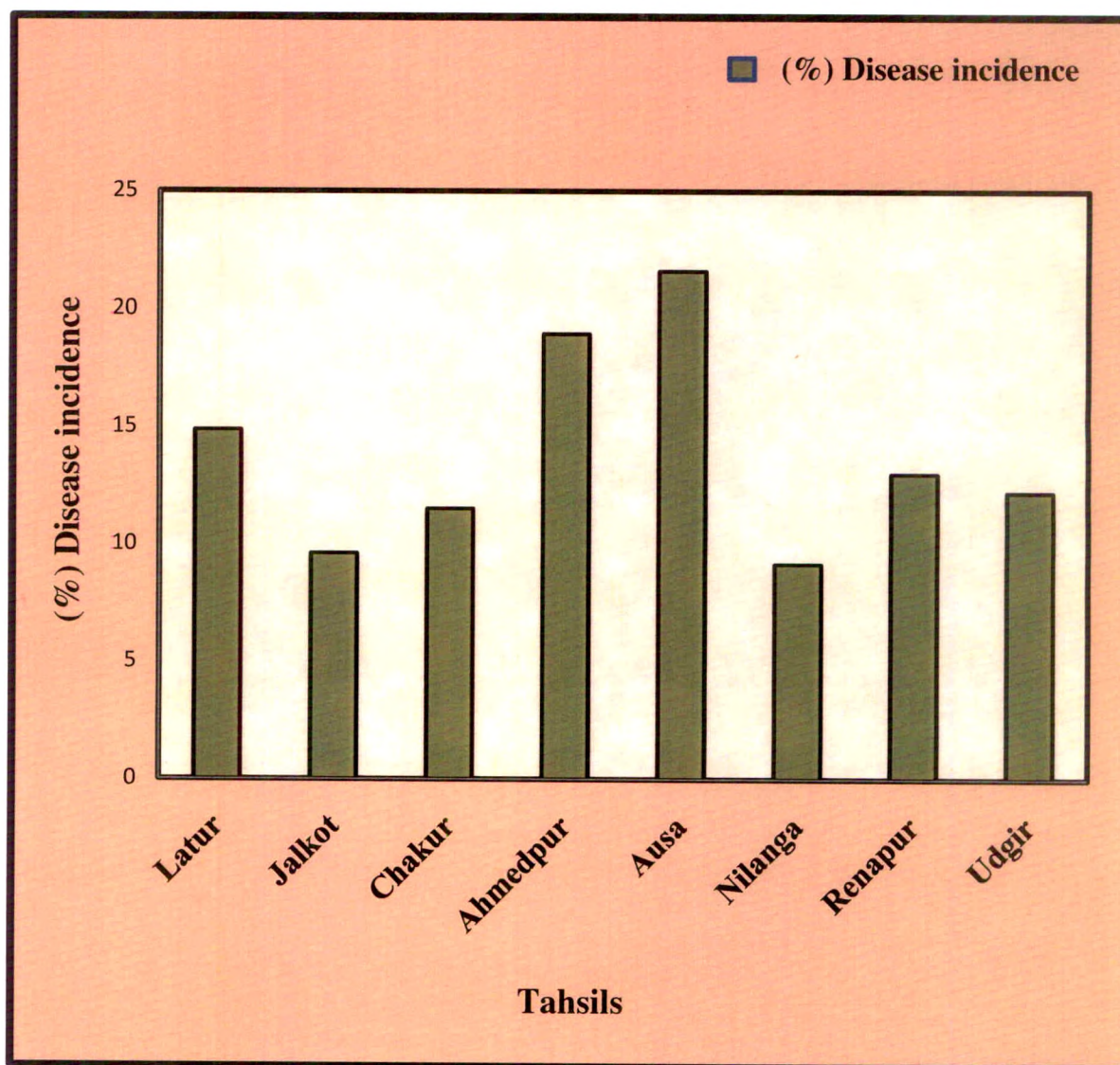


Fig.2 Incidence of Chickpea dry root rot on farmer's field in Latur district

PLATE - II



PLATE – II Differential growth of *Macrophomiona phaseolina* (Tassi)
Goid. Isolates.

From data presented in Table 1, it is revealed that heavy disease incidence was noticed in Ausa Tahsil (21.60 per cent) followed by Ahmedpur (18.92 per cent). Moderate disease incidence was recorded in tahsils Latur (14.21 per cent), Renapur (13.00 per cent). Udgir (12.20 per cent) and Chakur (11.50 per cent), whereas lowest dry root rot disease incidence was recorded in Jalkot (9.60 per cent) and Nilanga (9.15 per cent). The average dry root rot disease incidence recorded in Latur district was to the tune of (13.85 per cent).

Chickpea dry root rot samples from eight tahsils of Latur district were separately subjected to isolate on PDA medium in laboratory to find out the major pathogen associated with dry root rot on the basis of morphological and cultural characteristics developed on PDA medium, organism associated was confirmed by following Koch's postulates (Plate-II).

4.2 Isolation, Pathogenicity, Reisolation and Symptomatology

4.2.1 Isolation

The fungus *Macrophomina phaseolina* (Tassi) Goid was isolated from the dry root rot infected Chickpea plant samples collected during the survey of Latur district. The isolation was carried out on Potato dextrose agar (PDA) medium. *Macrophomina phaseolina* produced white, septate mycelial growth which later turned to dark grey (Plate-III-A) and produced abundant, dark brown sclerotia on PDA (Plate-III-B). Pure culture was obtained by sub culturing on PDA plates transferred and maintained on Potato dextrose agar (PDA) slants for further studies.

4.2.2 Pathogenicity Test

Pathogenicity of the test fungus (*Macrophomina phaseolina*) was proved by soil inoculation method in pot culture using susceptible cultivar JG-62. The pathogen *Macrophomina phaseolina* quickly

PLATE - III



PLATE – III - (A) Pure culture of *Macrophomiona phaseolina* (Tassi) Goid.



PLATE – III - (B) Sclerotia of *Macrophomiona phaseolina* (Tassi) Goid.

PLATE - IV

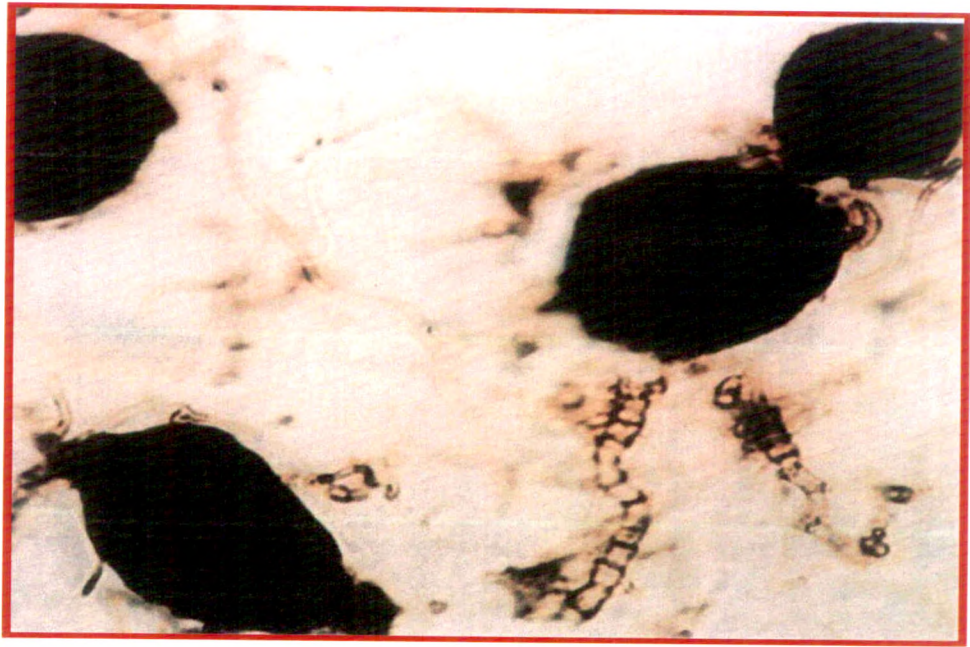


PLATE – IV Microphotograph of sclerotia of *Macrophomiona phaseolina* (Tassi) Goid

colonized the soil in pots within 10 days of inoculation. The typical dry root rot symptoms showing drying of plants started to develop 40 days after inoculation. Symptoms developed were found to be similar to that of naturally infected Chickpea plants.

Symptoms of dry root rot of Chickpea initiated at seedling stage. This seedling retained an almost green colour in the earlier stages which later appeared pale yellow in colour and then turned yellow. The leaves of such wilted and dried plants appeared to be straw coloured. Petioles and rachis showed dropping along with leaflets. This indicates that the *Macrophomina phaseolina* isolated obtained from naturally rotted plants could also cause similar infection in Chickpea producing typical dry root rot symptoms and the disease incidence was 100 per cent in sick soil. Whereas it was 00.00 per cent in control pot (Plate - V)

4.2.3 Reisolation

The fungus was reisolated on PDA from artificially inoculated plants showing typical dry root rot symptoms typical fungal growth of original culture.

4.2.4 Symptomatology

In artificially inoculated plants produced typical dry root rot symptoms in JG-62 a susceptible variety of Chickpea. Symptoms of dry root rot of Chickpea are initiated at seedling stages. The seedling retained an almost normal green colour in the earlier stage which later appeared pale yellow in colour and later turned yellow. In the advanced stage of the disease the plants started wilting from the base of the plant and by flowering to podding stage it had completely wilted. The leaves of such wilted and dried plants appeared straw coloured. Petioles and rachis showed dropping along with leaflets. Dried leaflets of affected plants did not shed at maturity.

PLATE - V



PLATE –V- (A) Sand maize media with culture of *Macrophomiona phaseolina* (Tassi) Goid.



Diseased

Healthy

PLATE –V- (B) Pathogenicity test of *Macrophomiona phaseolina* on Chickpea cultivar JG-62.

The affected plants could be easily pulled out as there was necrosis of roots and root proliferation was also observed to be reduced considerably. The barks near ground level get shredded. The blackish sclerotial bodies were observed on the exposed and inner side of the bark of the roots and particularly at collar region (Plate-VI). The affected roots appeared black in color and when these were cut open longitudinally (Plate-VII-A) and in the transverse section minute black sclerotial bodies in the pith near cortex were noticed (Plate-VII-B).

4.2.5 Identification of fungal pathogen

The pathogenic culture isolated from diseased plant was identified on the basis of morphological characters and was get confirmed from Agharkar Institute, Pune.

4.2.6. Cultural Studies

The cultural characteristics *viz.*, colony morphology, mycelial growth and sclerotial production of *Macrophomina phaseolina* were studied *in vitro* using various culture media. A total six culture media *viz.*, Oat meal agar, Richards agar, Asthana Hawkers media, Czapek dox agar, Host leaf extract agar, Potato dextrose agar were evaluated as detailed under 3.2.5.

PLATE - VI

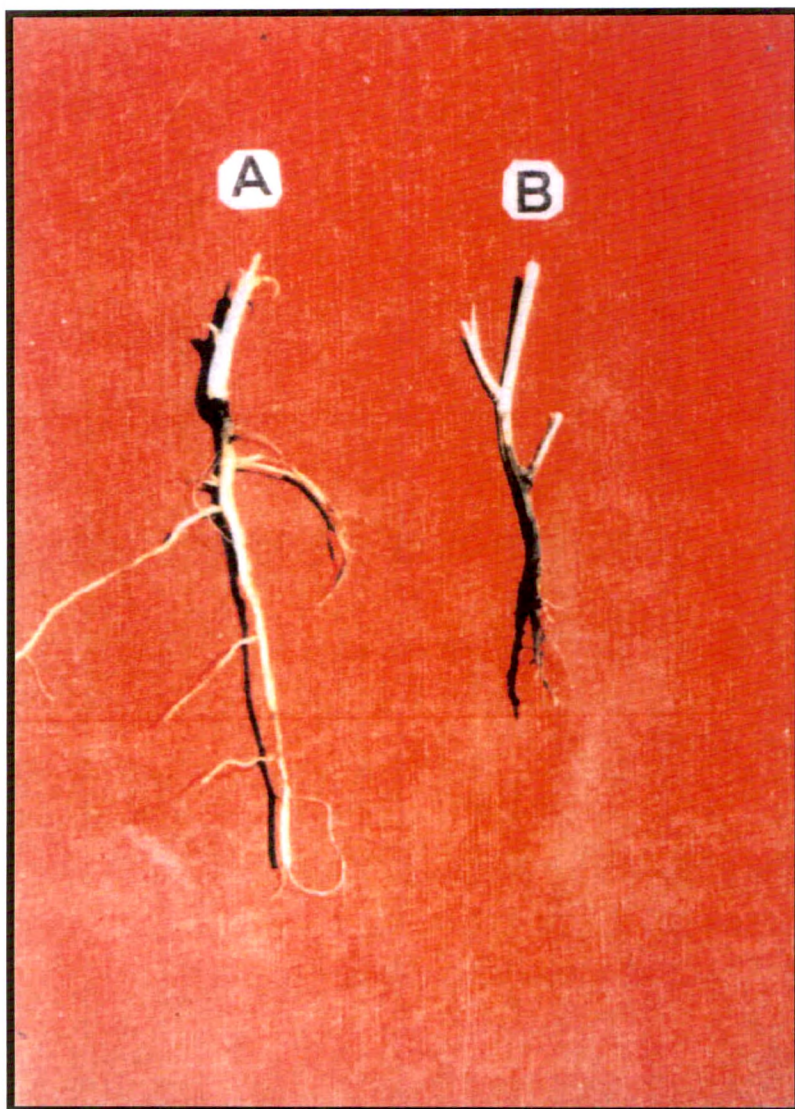


PLATE - VI Roots of chickpea infected by *Macrophomiona phaseolina* (Tassi) Goid showing blackening of roots and collar region.

A – Healthy root.

B – Diseased root.

PLATE - VII



PLATE -VII - (A) Infected root of Chickpea affected by *Macrophomiona phaseolina* (Tassi) Goid. Showing presence of sclerotia.

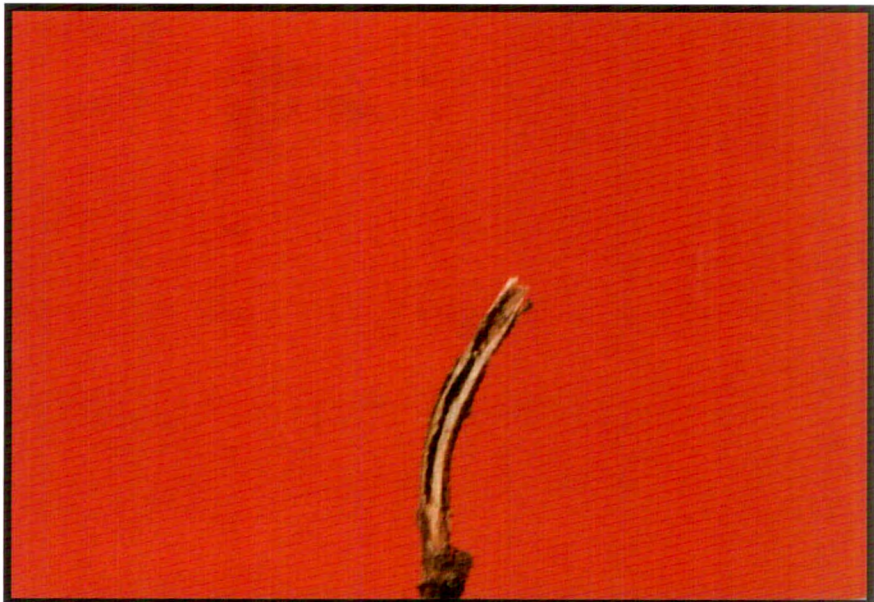


PLATE -VII- (B) Portion of root of Chickpea affected by *Macrophomiona phaseolina* (Tassi) Goid. Showing hallow blackened pith.

Table No. 2 Effect of different culture media on growth and sclerotial production of *Macrophomina phaseolina* isolate.

Sr. No.	Medium	Mean Colony Diameter (mm)*	Sclerotial production after 6 days
1.	Oat meal agar	73.53	+
2.	Richards agar	75.60	+++
3.	Asthana Hawakers Media	34.10	++
4.	Czapek's dox agar	78.26	+++
5.	Host leaf extract media	82.30	++++
6.	Potato dextrose agar	90.00	++++
	S.E. $\pm$	0.9	
	C.D. at 5%	2.87	

* = Average of three replication.

++++ = Excellent Sclerotial production.

+++ = Fair Sclerotial production.

++ = Good Sclerotial production.

+ = Poor Sclerotial production.

The results (Table -2) revealed that, among six culture media tested, potato dextrose agar was found most suitable which encouraged maximum radial mycelia growth (90.00 mm) (Plate-VIII and depicted in fig. 3). This was followed by Host leaf extract agar (82.30 mm), Czapek's dox agar (78.26 mm), Richards's agar (75.60 mm) and Oat meal agar (73.53mm) respectively. Least growth was observed on Asthana Hawakers media (34.10 mm).

The excellent sclerotial production was observed in Potato dextrose agar and host leaf extract agar, fair sclerotia production was observed in Czapek dox agar and Richards's agar, good sclerotia production was

PLATE - VIII

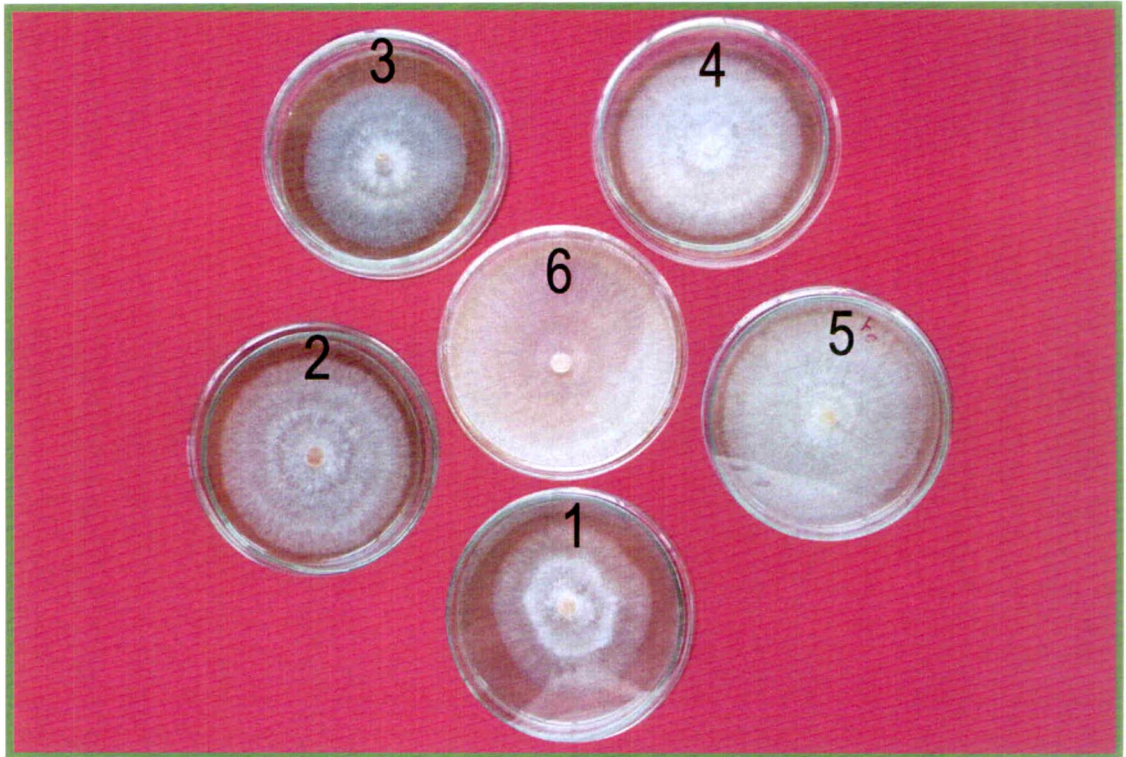


PLATE – VIII Growth of *Macrophomiona phaseolina* (Tassi) Goid. On different culture media.

- 1. Oat meal agar.**
 - 2. Richards's agar media.**
 - 3. Asthana Hawakers media.**
 - 4. Czapek's dextrose agar.**
 - 5. Host leaf extract agar.**
 - 6. Potato dextrose agar.**
-

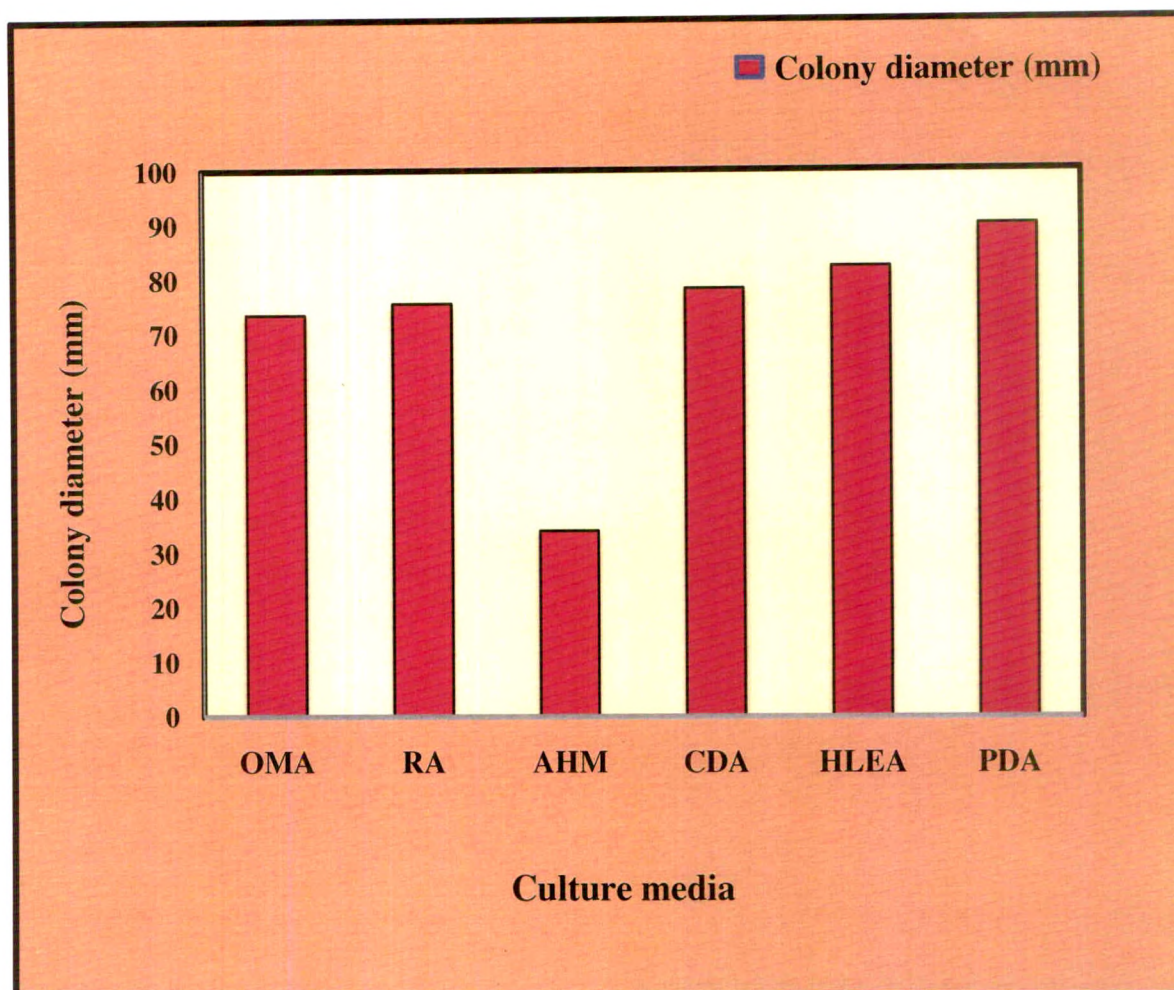


Fig.3 Growth of *Macrophomina phaseolina* (Tassi) Goid on different culture media.

OMA : Oat meal agar.

RA : Richard's agar.

AHM : Asthana and Hawakers media.

CDA : Czapek's dox agar.

HLEA : Host leaf extract agar.

PDA : Potato dextrose agar.

observed in Asthana Hawker's media and among these medium poor sclerotia production was observed in oat meal agar medium.

4.2.7. Cultural Characters

All the culture media tested exhibited wide range of variations in the colony characteristics. Observations on the colony characters are given in below. The colony characteristics viz., topography amount of mycelia growth and pigmentations of *Macrophomina phaseolina*. The test pathogen produced poor to abundant aerial mycelium with flat, submerged and scattered topography.

Table No. 3 Cultural characteristics of isolates on different culture media.

Sr. no.	Media	Cultural characteristics
1.	Oat meal agar	Mycelial growth flat, scattered and poor; colour initially white later become grayish dark after 6 days. Sclerotia appeared after 7 days of inoculation.
2.	Richards agar	Mycelium abundant, cottony, partial aerial and partially submerge; colour initially white later become brown after 5 days. Sclerotia appeared after 8 days of inoculation.
3.	Asthana Hawakers Media	Colony grey to dark flat scattered, sparse and poor growth.
4.	Czapek's dox agar	Mycelium aerial and abundant, partially wooly and cottony; initially white later changed greenish brown after 3 days and finally brown after 5 days. Sclerotia appeared after 6 days of inoculation.
5.	Host leaf extract media	Mycelial growth aerial, abundant and submerged; colour initially white, later changed to blackish brown. Sclerotia appeared after 6 days of inoculation.
6.	Potato dextrose agar	Mycelium abundant, aerial as well as submerged wooly; colour initially white later changed to dark brown after 5 days. Sclerotia appeared after 5 days of inoculation.

Maximum radial mycelia growth and excellent sclerotial production was observed on Potato dextrose agar. Hence for multiplication and maintenance of *Macrophomina phaseolina* (Tassi) Goid, Potato dextrose agar (PDA) medium was used. The most commonly observed pigmentations were white, cottony white which later upon ageing of the culture turned brown to black and grayish. The sclerotial production in all culture media tested initiated at 5-8 days after inoculation.

4.3 Pot Culture

The three fungicides, three bioagents were evaluated by seed treatments method and three organic soil amendments were evaluated by soil application method in pot culture experiment (as detailed under 3.3) with Chickpea cultivar JG-62 against *M. phaseolina*. Observations on seed germination, per cent disease incidence and percent disease control were recorded.

Table No.4 Evaluation of fungicides, bioagents and organic soil amendments against *Macrophomina phaseolina* with Chickpea cultivar, JG-62(Pot culture).

Sr. No	Treatment	Doses	Germination (%)*	Mean dry root rot disease incidence (%)*	Mean dry root rot disease Control (%)
1.	Thiram (75% WP)	(@3 gm/kg seed)	84.66 (57.88)	26.33 (15.26)	73.67
2.	Carbendazim (50 % WP)	(@2 gm/kg seed)	90.00 (64.23)	14.00 (8.04)	86.00
3.	Captan (50% WP)	(@3 gm/kg seed)	81.33 (54.44)	28.20 (16.38)	71.80
4.	<i>Trichoderma harzianum</i>	(@10 gm/kg seed)	84.99 (58.20)	35.33 (20.69)	64.67
5.	<i>Trichoderma viride</i>	(@10 gm/kg seed)	92.00 (67.03)	31.15 (18.15)	68.85
6.	<i>T. hamatum</i>	(@10 gm/kg seed)	84.50 (57.68)	43.30 (25.65)	56.70
7.	Neem cake	(@10 gm/kg soil)	85.66 (58.93)	14.70 (8.45)	85.30
8.	Cotton cake	(@10 gm/kg soil)	81.66 (54.09)	21.25 (12.27)	78.75
9.	Groundnut cake	(@10gm/kg soil)	82.33 (56.14)	24.50 (14.18)	75.50
10.	Control (Untreated)		74.66 (48.30)	100 (89.99)	
	S.E.+		1.13	0.60	
	C.D. at 5%		3.39	1.79	

Figures in parentheses are arc sine transformed values.

* = Average of three replication

Result (Table-4) revealed that fungicides, bioagents and organic soil amendments tested were found effective against the test pathogen and significantly enhanced seed germination and reduced the per cent disease incidence in Chickpea over untreated control (Plate –IX and depicted in fig.4.). In all treatments fungicides, bioagents and organic soil

PLATE - IX



PLATE – IX Evaluation of fungicides, bioagents and organic soil amendments against *Macrophomiona phaseolina* (Tassi) Goid. With Chickpea cultivar JG-62 (Pot culture).

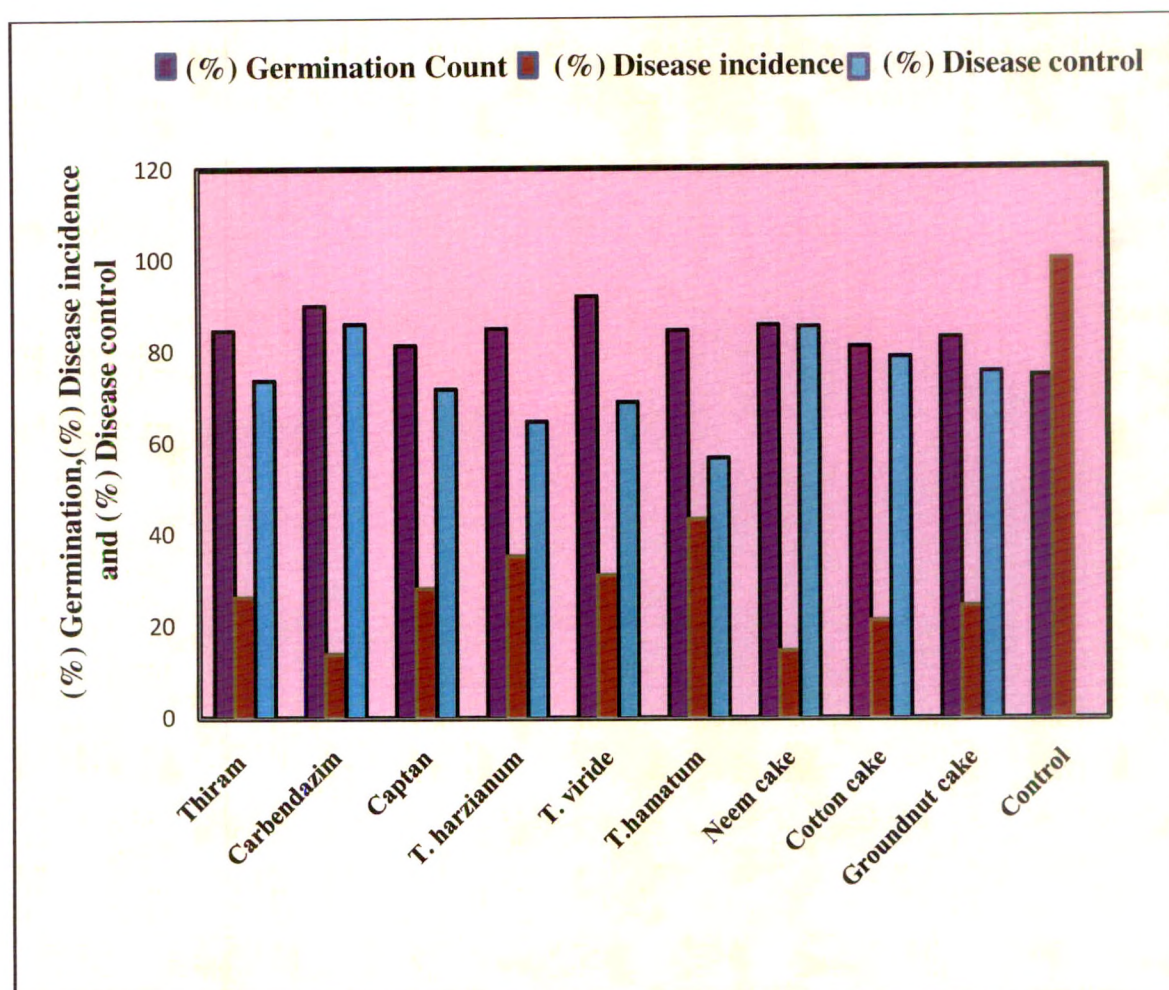


Fig.4 Evaluation of fungicides, bioagents and organic soil amendments on germination, disease incidence and disease control against *M.phaseolina* with Chickpea Cv.JG-62 (Pot culture).

92 per cent as compared to control (74.66 per cent). However, maximum per cent germination was recorded in the treatment *T. viride* (92.00 per cent). This was followed by Carbendazim (90.00 per cent), Neem cake (85.66 per cent), *T. harzianum* (84.99 per cent), Thirum (84.66 per cent), *T. hamatum* (84.50 per cent), Groundnut cake (82.33 per cent), Cotton cake (81.66 per cent) and Captan (81.33 per cent) as compared to untreated control (74.66 per cent).

In least disease incidence was recorded in treatment Carbendazim (14.00 per cent) and this was followed by Neem cake (14.70 per cent) and Cotton cake with (21.25 per cent) and Groundnut cake with (24.50 per cent) and Thirum with (26.33 per cent) and Captan (28.20 per cent) and *T. viride* with (31.15 per cent) and *T. harzianum* with (35.33 per cent) and *T. hamatum* with (43.30 per cent). Over control which showed maximum disease incidence (100 per cent).

In the average disease control in all the treatment significantly controlled disease incidence over control. Carbendazim gave maximum disease control (86.00 per cent), followed by Neem cake (85.30 per cent) and Cotton cake (78.75 per cent) and Groundnut cake (75.50 per cent) and Thirum (73.67 per cent) and Captan (71.80 per cent) and *T. viride* (68.85 per cent) and *T. harzianum* (64.67 per cent) and *T. hamatum* (56.70 per cent) over control (00.00 per cent).

Among three groups in chemical group Carbendazim was found best which gave maximum germination count (90.00 per cent) with average dry root rot incidence (14.00 per cent) and maximum disease control (86.00 per cent) as compared to Thirum and Captan over control. In biocontrol group *T. viride* was found most effective as compared to other two bioagents (i.e. *T. harzianum* and *T. hamatum*) and *T. viride* gave maximum germination count with (92.00 per cent) with disease incidence (31.15 per cent) and disease control (68.85 per cent) over

control. In organic soil amendments Neem cake was found most effective as compared to Groundnut cake and Cotton cake. Neem cake gave maximum germination count (85.66 per cent) with least disease incidence (14.70 per cent) and disease control (85.30 per cent).

4.4. Field evaluations

Integrated management of Chickpea dry root rot disease with fungicides and bioagents were evaluated against *Macrophomina phaseolina* in field condition by applying RBD and using susceptible Chickpea cultivar JG-62. As detailed under 3.4.1.

Table No.5 Field evaluation of fungicides and bioagents against *Macrophomina phaseolina* (Tassi) Goid in Chickpea cultivar JG-62.

Sr. No.	Treatment Details	Mean dry root rot disease incidence (%)*	Meandry root rot disease Control (%)	Average yield (Kg/ha)	Test Weight (gm)
1	Thiram (75% WP)	13.20 (7.58)	80.19	1800	15.53
2	Carbendazim (50%WP)	15.60 (8.97)	76.59	1625	15.50
3	Carbendazim(50%W P)+Thiram(75% WP)	10.86 (6.23)	83.70	1950	16.36
4	Carbendazim (50%WP)+Mancozeb (50% WP)	30.20 (17.57)	54.69	1291	14.36
5	<i>T. harzianum</i>	22.48 (12.99)	66.27	1466	14.79
6	<i>T. viride</i>	20.66 (12.99)	69.00	1566	15.12
7	<i>Pseudomonas fluroescens</i>	26.76 (15.52)	59.85	1416	14.52
8	<i>T. harzianum</i> + <i>T. viride</i>	12.30 (8.22)	81.54	1825	16.02
9	Control (untreated)	66.66 (41.82)		1170	11.67
	S.E.+	0.69		1.33	0.8
	C.D. at 5%	2.07		4.00	2.4

Figures in parentheses are arc sine transformed values.

* = Average of three replications.

Result represented in Table 5 revealed that all seed treatments with fungicides and bioagents significantly controlled dry root rot disease incidence and increased seed yield and test weight in Chickpea cultivar JG-62. (Plate-X and depicted in fig. 5, 6 and 7). However the treatment no.3 i.e. Carbendazim + Thiram was found superior over all treatments and recorded significantly least mean disease incidence (10.86 per cent)

PLATE - X



**PLATE – X Field evaluation of fungicides and bioagents
against *Macrophomiona phaseolina* (Tassi) Goid. With
Chickpea cultivar JG-62.**

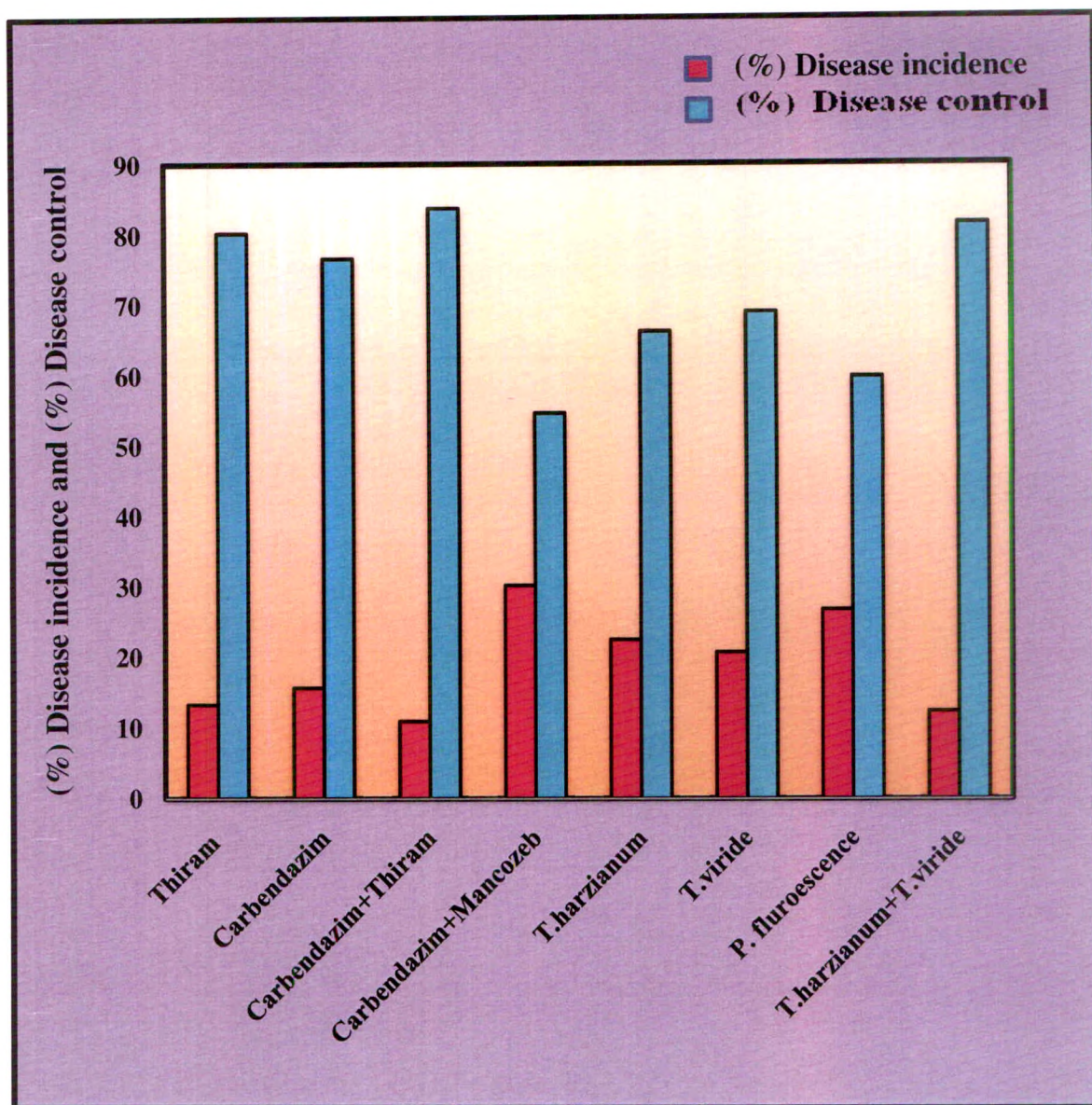


Fig.5 Field evaluation of fungicides and bioagents on disease incidence and disease control against *M.phaseolina*.

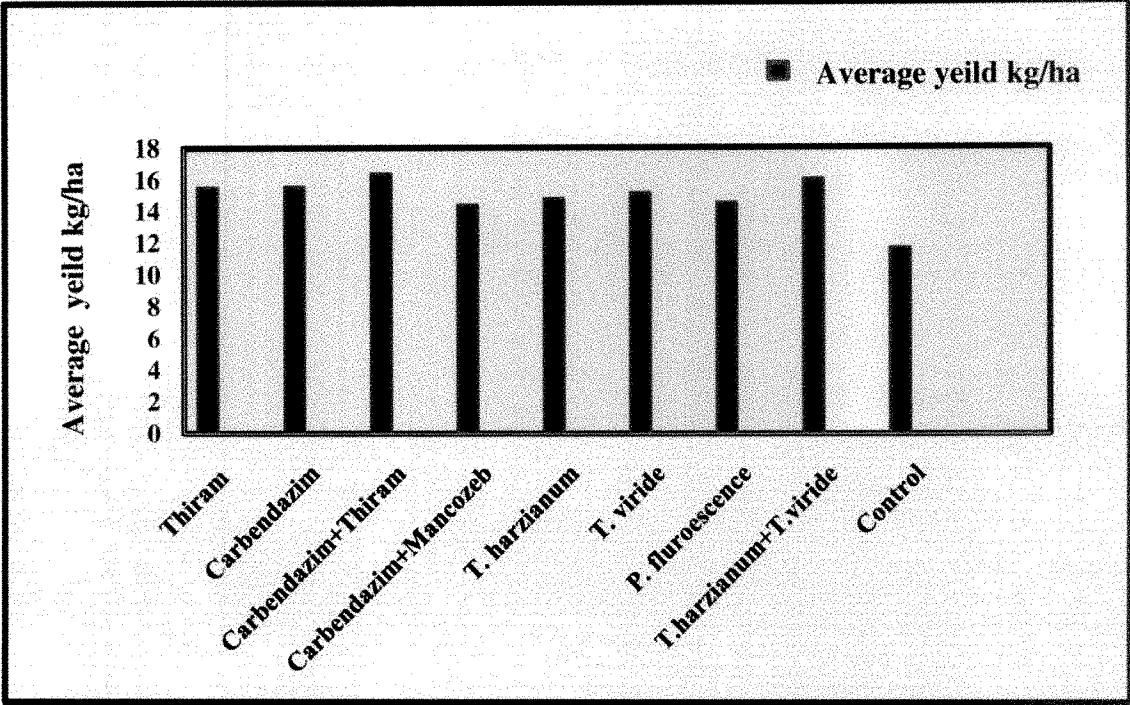


Fig.6 Field evaluation of fungicides and bioagents on average yield against *M.phaseolina*.

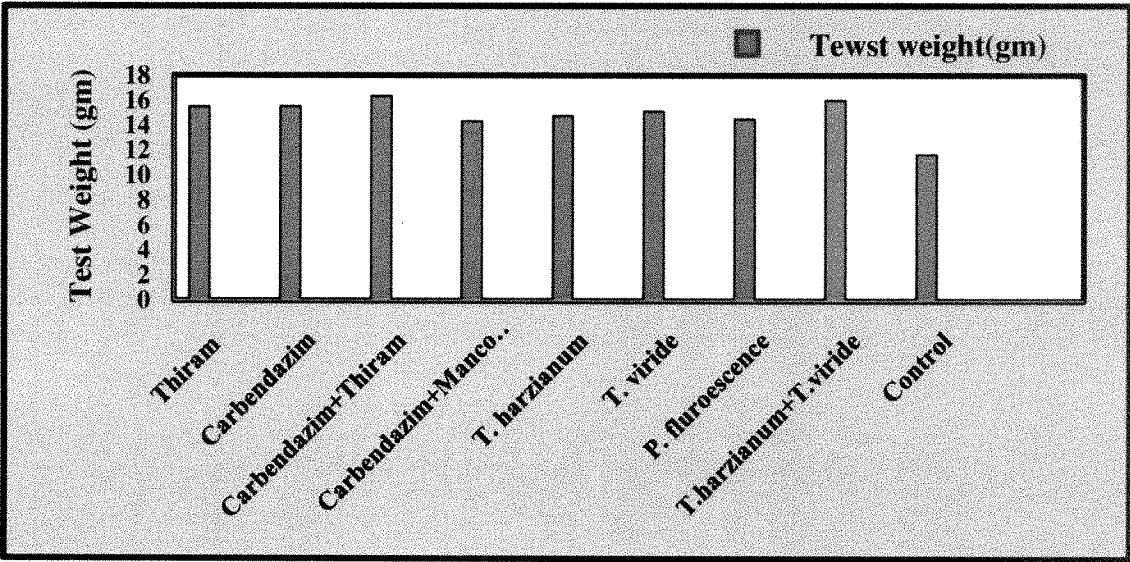


Fig.7 Field evaluation of fungicides and bioagents on test weight against *M.phaseolina*.

(1950 kg/ha) and test weight (16.36 gm). This was followed by treatment no.8 i.e. *T. harzianum* + *T. viride* which recorded second least mean disease incidence (12.30 per cent) with maximum disease control (81.54 per cent) with highest average seed yield (1825 kg/ha) and test weight (16.02gm). This was followed by the treatment no. 1 i.e. alone treatment Thirum which recorded least disease incidence (13.20 per cent) with maximum disease control (80.19 per cent) with highest average seed yield (1800 kg/ha) and test weight (15.53 gm), followed by Carbendazim which recorded mean disease incidence (15.60 per cent) with disease controls (76.59 per cent) with average seed yield (1625 kg/ha) and test weight (15.50 gm), followed by the treatment no.6 i.e. *T. viride* with disease incidence (20.66 per cent) and disease control (69.00 per cent) with seed yield (1566 kg/ha) and test weight (15.12 gm) and treatment no.5 i.e. *T. harzianum* with disease incidence (22.48 per cent) and disease control (66.27 per cent) with yield (1466 kg/ha) and test weight (14.79 gm) and treatment no.7 i.e. *P. fluroescens* with disease incidence (26.76 per cent) and disease control (59.85 per cent) with seed yield (1416 kg/ha) and test weight (14.52 gm) and treatment no.4 i.e. combination of Carbendazim 50 % WP + Mancozeb 50% WP with disease incidence (30.20 per cent) and disease control (54.69 per cent) with average seed yield (1291 kg/ha) and test weight (14.36 gm). All these treatments significantly controls disease incidence over control and increased seed yield and test weight.

In the chemical group, treatment no. 3 i.e. Carbendazim 50% WP +Thirum 75 %WP found superior over all the chemical treatment with least mean incidence (10.86 per cent) with maximum disease controls (83.70 per cent) with highest seed yield (1950 kg/ha) and test weight (16.36 gm). This was followed by the treatment no. 1 i.e. alone treatment Thirum 50 %WP which recorded least disease incidence (13.20 per cent)

with maximum disease control (80.19 per cent) with highest average seed yield (1800 kg/ha) and test weight (15.53 gm).

In biocontrol group the treatment no.8 i.e. *T. harzianum* + *T. viride* found superior over all the biological treatment with least mean disease incidence (12.30 per cent) with maximum disease control (81.54 per cent) with highest average seed yield (1825 kg/ha) and test weight (16.02gm).

Among the two treatments group viz., chemical and biocontrol, the treatment no. 3 i.e. Carbendazim 50% WP +Thirum 75 % WP was found superior in controlling disease incidence with average percent with average disease incidence (83.70 per cent) and (10.86 per cent) with highest seed yield (1950 kg/ha) and test weight (16.36 gm). This was followed by treatment no.8 i.e. *T. harzianum* + *T. viride* in controlling disease incidence with average per cent disease incidence (81.54 per cent) and (12.30 per cent) respectively with highest seed yield (1825 kg/ha) and test weight (16.02 gm).

4.5. Varietal Screening of Chickpea against *M. Phaseolina* in pot culture.

The screening of 20 Chickpea varieties was done by artificial inoculation with *M.phaseolina* in pot culture (Plate –XI) on the basis of per cent disease incidence, germplasm lines, cultivars were categorized as resistant, moderately resistant, susceptible and highly susceptible, adopting disease rating scale (Nene *et al.* 1981). The result obtained on percent disease incidence of different germplasm lines evaluated for the *Macrophomina phaseolina* interreaction in sick pot developed at department of Plant Pathology, College of Agriculture, Latur are presented in Table no.6 and (Plate -XI).

PLATE - XI



PLATE – XI Varietal screening of germplasm lines of chickpea against *Macrophomiona phaseolina* (Tassi) Goid.

Table No.6 Reaction of Chickpea germplasm lines against *M. phaseolina*

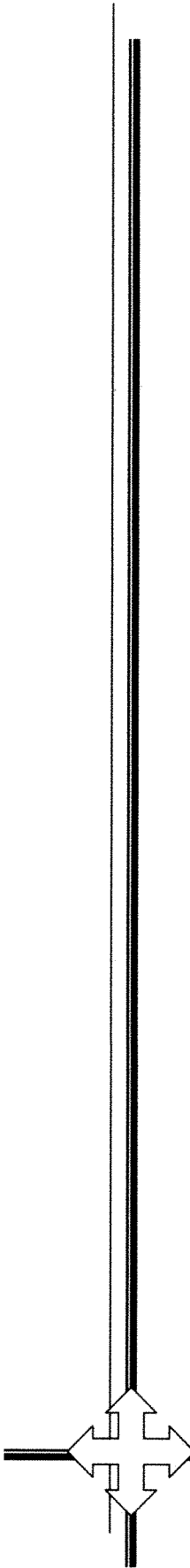
Sr. No.	Name of genotype	Mean percent disease incidence(%) *	Disease reaction
1.	Vishal	12.50	MR
2.	BCP-73	7.23	R
3.	PG-12	1.23	R
4.	Virat	12.26	MR
5.	Jaki-9218	6.13	R
6.	BCP-3	4.17	R
7.	PG-97022	57.32	HS
8.	PG-96329	13.78	MR
9.	BDNG-2002	99.00	HS
10.	PG-97313	58.62	HS
11.	PG-9425-5	6.94	R
12.	BCP-107	2.10	R
13.	PG-9757-6-1	23.16	MR
14.	PG-9557-2-1	9.61	R
15.	PG-9758-6-2	2.87	R
16.	BCP-15	8.24	R
17.	PG-94094	32.60	S
18.	PG-98902	98.21	HS
19.	BCP-28	8.33	R
20.	PG-9755-8-1	25.47	S

* : Average of two replications.

R : Resistant. MR : Moderately Resistant

S : Susceptible. HS : Highly Susceptible.

Amongst 20 germplasm lines 10 lines viz., BCP-73, PG-12 , Jaki-9218, BCP-3, PG-9425-5, BCP-107, PG-9557-2-1, PG-9758-6-2, BCP-15 and BCP-28 were found to be resistant with an average disease incidence (6.88 per cent), 4 lines viz., Vishal , Virat, PG-96329 and PG-9757-6-1 were found to be moderately resistant with average incidence (15.24 per cent), 2 lines viz., PG-94094 and PG-9755-8-1 were found to be susceptible with average disease incidence (29.03 per cent) and 4 lines viz., PG-97022, BDNG-2002, PG-97313, PG-98902, and were found to be highly susceptible with average disease incidence (78.28 per cent).



DISCUSSION



Chapter –V

Discussion

Chickpea (*Cicer arietinum* L.) is the world's third most important crop, which is also one of the important *Rabi* pulse crop grown in Maharashtra. Fifty different pathogens have so far been reported on Chickpea (Nene *et al.* 1989) including diseases caused by fungal, bacterial, nematodes, mycoplasma and viral pathogens. Several diseases reported from all growing states in India *viz.*, Punjab, Delhi, Madhya Pradesh, Bangladesh, Bihar, Tamilnadu, Uttar Pradesh, Karnataka, Gujarat, Maharashtra and U.S.A. (Nene *et al.* 1989).

Amongst all the diseases of Chickpea dry root rot caused by *Macrophomina phaseolina* (Tassi) Goid is predominant one and responsible for causing heavy crop losses. The pathogen has been considered of great economic importance because earlier, the pathogen has been reported to causes complex disease syndromes like seedling rot, root rot, ashy stem blight, collar rot, stalk rot, leaf spot and blights *etc.* (Saksena, 1979).

Keeping in view, economic importance of the crop and losses caused by dry root rot (*Macrophomina phaseolina*) being soil born, is difficult to control using chemicals only therefore integrated disease management of Chickpea dry root rot using resistant varieties, bioagents, fungicides and soil amendments would be reliable and most economical for successful cultivation of Chickpea. Considering the increasing incidence and spread of Chickpea dry root rot and yield losses caused thereby, the present investigations were undertaken on the following aspects *viz.* Survey on dry root rot of Chickpea, isolation, identification and pathogenicity and cultural characters of pathogen, *in vitro* evaluation of fungicides, bioagents and organic soil amendments, *in vivo* evaluation

of fungicides and bioagents and germplasm screening, at the department of Plant Pathology, College of Agriculture Latur.

The results obtained on various aspects during present investigations are being discussed here in the following paragraphs.

5.1. Survey

Survey was undertaken for recording incidence of Chickpea dry root rot complex in Latur district of Marathwada region. Maximum disease incidence was noticed in Ausa tahsils (21.60 per cent) followed by Ahmedpur (18.92 per cent) and minimum in Nilanga tahsil (9.15 per cent) and Jalkot tahsils (9.60 per cent) and Table 1.

In all locations in 8 tahsils of Latur district were traversed and recorded average disease incidence to the tune of (13.85 per cent). The results on the incidence of dry root rot disease are similar with the results to those of Sharma *et al.* (1983) they have reported that the mean dry root rot incidence of Chickpea ranged from 15.90 to 37.99 per cent and dry root rot incidence ranged from 3.58 to 20.63 per cent. Prajapati *et al.* (2004). They have reported that the incidence of dry root rot of Chickpea in different localities varied from 19.0 to 42.0 per cent.

5.2. Isolation, Pathogenicity, Reisolation and Symptomatology

5.2.1. Isolation

The pathogen, *Macrophomina phaseolina* was isolated on the on the basal medium Potato dextrose agar from the Chickpea plants showing typical symptoms of dry root rot. The test pathogen produced moderate to abundant; aerial to submerged mycelial growth on Potato dextrose agar (PDA). Initially the mycelium produced was white coloured which later turned dark grey. The pathogen also produced moderate to abundant dark brown coloured sclerotia within 5 to 8 days after inoculation.

The pure culture of the pathogen was identified and confirmed as *Macrophomina phaseolina* from, Agarkar Research Institute. The result

of present investigations correlates with finding of Bhatti *et al.* (1987), Bhate (1983).

5.2.2. Pathogenicity

In order to prove pathogenicity the sterilized soil was artificially inoculated with *Macrophomina phaseolina* (Tassi) Goid. The seeds of Chickpea susceptible cultivar (JG-62) were sown in pot and the plants were observed for the disease development. The plants produced symptoms which were identical to those observed in field condition.

Isolation identification and pathogenicity of *Macrophomina phaseolina* (*R. bataticola*) were attempted by several workers Bhate (1983), isolated and proved pathogenicity of *Macrophomina phaseolina* on Green gram Vishwadhara and Sarbhoy (1991), *R. bataticola* (*Macrophomina phaseolina*) isolates and proved pathogenicity on Soybean plants, Patel and Anahosur (2000) found isolated and proved pathogenicity of soil born fungi, *Fusarium spp.*, *Macrophomina phaseolina* and *Sclerotium rolfsii* in Chickpea, Katariya *et al.* (2007), isolated and proved pathogenicity of *R. bataticola* (*Macrophomina phaseolina*) in Chickpea plants.

5.2.3. Reisolation

Reisolation studies revealed that the presence of same fungus identified from naturally infected plants. The morphological and cultural characteristics of *Macrophomina phaseolina* obtained after reisolation were similar to those reported earlier by workers Bainde *et al.* (1978), Goyal and Mehrotra (1982) and Sobti and Bansal (1988).

5.2.4. Symptomatology

Pure culture of *Macrophomina phaseolina* was again reinoculated on Chickpea cultivar JG-62 in pot culture and the symptoms appeared were initially on wilting retain green colour which later turn yellow and

infected plants suddenly dried and wilted. Roots appeared black in colour and histopathologically there was presence of minute black sclerotia.

These symptoms are in conformity with those described by Vishwadhar and Sarbhoy (1991), Bhate (1983), Patel and Anahosur (2000), Singh *et al.* (1990), Nene *et al.* (1978),

5.2.5. Cultural Character

A total six culture media viz., Oat meal agar, Richards's agar, Asthana Hawakers media, Czapek's dox agar, Host leaf extract agar and Potato dextrose agar were used to study their effect on cultural and morphological characteristics and sclerotial production of *Macrophomina phaseolina*. The result obtained (Table -2) indicated that all the culture media tested exhibited near about similar effects on the cultural and morphological characteristics of the test pathogen. However, Potato dextrose agar favored maximum radial growth (90.00 mm), followed by Host leaf extracts (82.30 mm), Czapek's dox agar (78.26 mm), Richards's agar (75.60 mm) and produced excellent to fair sclerotia production. Rest of the culture media tested also exhibited mycelia growth and sclerotia production of the test pathogen. The amount of mycelium produced and sclerotial production were excellent to poor on all the culture media tested.

The result obtained from Table-3 indicated that topography of the colonies was flat to submerged, pigmentations were white, cottony white to brown black and grayish. Result of the present study on effect of various culture media on mycelial growth and sclerotial production of *Macrophomina phaseolina* (*R. bataticola*) and pigmentation are corresponds of with the results of those reported by earlier workers. Biande *et al.* (1978), Goyal and Mehrotra (1982), Sobti and Bansal (1988), Kulkarni *et al.* (1992), Surichandraselvan and Seetharaman (2003) and Salunkhe *et al.* (2009).

5.3. Evaluation of fungicides, bioagents and organic soil amendments against *Macrophomina phaseolina* (Pot culture)

Integrated management of Chickpea dry root rot disease with fungicides, bioagents and organic soil amendments were evaluated against *Macrophomina phaseolina* with Chickpea cultivar JG-62 in pot culture. All the treatment significantly enhanced seed germination and least disease incidence with maximum disease control over control treatment.

The percentage of seed germination recorded with the all fungicides, bioagents and organic soil amendments were ranged from (81.66 to 92.00 per cent) over untreated control (74.66 per cent). However, the highest seed germination was recorded with *T. viride* (92.00 per cent) followed by Carbendazim (90.00 per cent), Neem cake (85.66 per cent), *T. harzianum* (84.99 per cent), Thiram (84.66 per cent), *T. hamatum* (84.50 per cent), Groundnut cake (82.99 per cent), Cotton cake (81.66 per cent) and Captan (81.33 per cent) over untreated control (74.66 per cent).

The disease incidence in all the treatments was ranged from (14.00 to 43.30 per cent). The least disease incidence was recorded in Carbendazim 50 %WP (14.00 per cent). This was followed by Neem cake (14.70 per cent) followed by Cotton cake (21.25 per cent), Groundnut cake (24.50 per cent), Thiram 75 %WP (26.33 per cent), and Captan (28.20 per cent), *T. viride* (31.15 per cent), *T. harzianum* (35.33 per cent), *T. hamatum* (43.30 per cent) over untreated control which recorded maximum disease incidence (100.00 per cent).

In the disease control all the treatment significantly controls disease over untreated control. The maximum disease control was ranged from (56.70 to 86.00 per cent). The maximum disease control recorded with Carbendazim (86.00 per cent). This was followed by Neem cake

(85.30 per cent), Cotton cake (78.75 per cent), Groundnut cake (75.50 per cent), Thiram (73.67 per cent), Captan (71.80 per cent), *T. viride* (68.85 per cent), *T. harzianum* (64.67 per cent) and *T. hamatum* (56.70 per cent) over the control treatment which recorded (00.00 per cent) disease control.

The results obtained from present studies are similar with the results of earlier workers. Singh *et al.* (1987), Taya *et al.* (1990), Kehri and Chandra (1991), Sankar and Jeyarajan (1996), Jakhar *et al.* (1997), Hundekar *et al.* (1998), Singh *et al.* (1998), Theradimani and Hepziba (2003), Sharma and Gupta (2004), Sajeena *et al.* (2004), Surichandraselvan (2005), Vijay-Mohan *et al.* (2006), Lokesha and Benagi (2007), Mallesh *et al.* (2008), Sudha and Prabhu (2008), Kaushal and Sood (2008), Khan and Gangopadhyay (2008), Jaiman *et al.* (2009) and Choudhary *et al.* (2010).

5.4 Field evaluation of fungicides and bioagents against *Macrophomina phaseolina* in Chickpea cultivar JG-62.

Integrated management of Chickpea dry root rot disease with fungicides and bioagents were evaluated against *Macrophomina phaseolina* in field condition applying RBD by using Chickpea cultivar JG-62 as detailed under 3.4.1.

Result of field studies (Table-5) revealed that all the seed treatments with fungicides and bioagents alone and their combinations significantly controlled the disease incidence and increased the seed yield as well as test weight in Chickpea cultivar JG-62. However the seed treatment with fungicides Carbendazim + Thiram and seed treatment with bioagents *T. harzianum* + *T. viride* were recorded significantly least disease incidence (10.86 per cent) and (12.30 per cent) and maximum disease control (83.70 per cent) and (81.54 per cent) respectively over untreated control. Both the seed treatment recorded seed yield and test

weight (1950 kg/ha) and (1825 kg/ha) and (16.36 gm) and (16.02gm) respectively over untreated control. However in fungicidal seed treatment alone Thiram and Carbendazim were recorded disease incidence (13.20 per cent) and (15.60 per cent) and disease controls (80.19 per cent) and (76.59 per cent) and average seed yield and test weight (1800kg/ha) and (1625kg/ha) and (15.53gm) and (15.50gm) respectively. Carbendazim + Mancozeb were least effective as compared to other treatments over control. This was followed by bioagents like *T viride*, *T. harzianum* and *P. fluroescens* which recorded disease incidence (20.66, 22.48 and 26.76 per cent) respectively and disease control (69, 66.27 and 59.85 per cent) and increased seed yield and test weight (1566, 1466 and 1416 kg/ha) and (15.12, 14.79 and 14.52 gm) over control (untreated) treatment.

Results of the present study on efficacy of fungicides and bioagents in controlling / reducing dry root rot in Chickpea are in conformity with those reported earlier by several workers. Sinha and khare (1977), Saxena and Saxena (1995), Rathore and Rathore (1999), Prajapati *et al.* (2004), Indra and Subbiah (2003), Surichandraselvan (2004), Jaiman and Jain (2004), Gaur et al.(2005), Sundravadana and Alice (2006), Bhattacharyya and Ghosh (2006), Vanitha and Alice (2008).

5.5. Varietal screening of Chickpea against *Macrophomina phaseolina* (Tassi) Goid

In all 20 lines / Varieties / cultivars were screened against *Macrophomina phaseolina*. Results revealed that 10 lines were found to be resistant, 4 lines as moderately resistant, 2 lines were found to be susceptible and 4 lines were found to be highly susceptible. The result on Chickpea varietal reaction against *Macrophomina phaseolina* recorded in present studies correlates with the results of earlier workers Reddy *et al.* (1991), Baker and Ahmed (1991), Zote *et al.* (1993), Prajapati *et al.*

(2003), Pande *et al.*(2004), Mohmad *et al.* (2005),Vijay-Mohan *et al.* (2006), Katariya and Gaur (2007).

SUMMARY

Chapter VI

Summary

Chickpea (*Cicer arietinum* L.) is leguminous crop popularly grown in India and other countries like China, Africa, South and North America etc. Chickpea (*Cicer arietinum* L.) is the world's third most important crop, and also in Maharashtra. The production and productivity of Chickpea in the state is comparatively low due to adverse effect of several biotic and abiotic factors. Amongst biotic factors fungus *Macrophomina phaseolina* (Tassi) Goid (*Rhizoctonia bataticola* (Taub) Britou Jones); causing dry root rot of disease plays an important role in decreasing production and productivity in the state. The fungus being soil born can survive for several of susceptible host; therefore it is very difficult to combat the disease by any one control measures.

Keeping in view, economic importance of the crop and losses caused by dry root rot (*Macrophomina phaseolina*) in Chickpea . Considering the increasing incidence and spread of Chickpea dry root rot and yield losses caused thereby, the present investigations were undertaken on the following aspects. viz, Survey on dry root rot of Chickpea, isolation, identification and pathogenicity and cultural characters of pathogen, evaluation of fungicides, bioagents and organic soil amendments in pot culture, *in vivo* evaluation of fungicides and bioagents and germplasm screening were undertaken during *rabi* 2010 at the department of Plant Pathology, College of Agriculture Latur. The results obtained on these aspects are summarized in the following paragraphs.

Survey of Chickpea dry root rot in Latur district of Marathwada region revealed that average Chickpea dry root rot incidence was to the tune of (13.85 per cent). Maximum disease incidence was noticed in

Ausa tahsils (21.60 per cent) followed by Ahmedpur (18.92 per cent) and minimum in Jalkot tahsils (9.60 per cent) and Nilanga tahsils (9.15 per cent) in Table 4.1.

Typical symptoms of the disease were developed in the artificially inoculated sick soil. On the basis of morphological characters, the organism was identified as *Macrophomina phaseolina* (Tassi) Goid.

The fungal pathogen produced dry root rot symptom on Chickpea plant was successfully isolated from roots on the basal culture medium Potato dextrose agar. Initially white coloured mycelium which later turned dark grey. The pathogen also produced moderate to abundant dark brown coloured sclerotia. The microscopic examination revealed that mycelium was closely septate and sclerotia observed were dark grey to brown and irregular to round in shape. Based on the symptomatology, morphological and cultural characteristics and microscopic observations, the test pathogen was identified and confirmed as *Macrophomina phaseolina* was proved *in vitro* on susceptible Chickpea cultivar JG-62.

A total six culture media were evaluated for their suitability for mycelial growth and sclerotial production of *Macrophomina phaseolina*. Out of six culture media tested, Potato dextrose agar encouraged significantly highest radial mycelia growth (90.00 mm) and excellent sclerotial production. This was followed by Host leaf extract agar with radial growth (82.30 mm) and excellent sclerotia production and Czapek's dox agar, Richards's agar which recorded radial mycelial growth 78.26 and 75.60 mm respectively and fair sclerotia production. Rest of the culture media tested exhibited moderate mycelial growth and sclerotia production. All the culture media exhibited a wide array of variations in the cultural characteristics of *Macrophomina phaseolina*.

In pot culture experiment result obtained in integrated disease management on *Macrophomina phaseolina* the seed treatments were

found effective against *Macrophomina phaseolina* and gave increased seed germination, reduced disease incidence and maximum disease control in Chickpea cultivar JG-62 over untreated control. However seed treatment with *T.viride* increased seed germination up to (92.00 per cent) followed by seed treatment with Carbendazim (90.00 per cent) and Neem cake (85.66 per cent) over control (74.66 per cent).

Among three group of seed treatment chemical treatment with Carbendazim was found best for increased seed germination up to (90.00 per cent), reduced disease incidence (14.00 per cent), and gave maximum disease control (86.00 per cent). In biological seed treatment group *T.viride* was found best for highest seed germination (92.00 per cent). Disease incidence (31.15 per cent) and disease control (68.85 per cent). In organic soil amendments group, Neem cake was recorded highest seed germination maximum (85.66 per cent) with least disease incidence (14.70 per cent) and maximum disease control (85.30 per cent).

In vivo (field) experiment, integrated management of Chickpea dry root rot disease fungicides and bioagents were evaluated against *Macrophomina phaseolina* significantly reduced per cent mean disease incidence, increased seed yield and test weight in Chickpea cultivar JG-62. Seed treatment with Carbendazim + Thiram significantly controlled disease incidence (83.70 per cent) with average disease incidence (10.86 per cent) with average seed yield 1950 kg/ha and test weight (16.36 gm). This was followed by *T. harzianum* + *T. viride* with control disease incidence (81.54 per cent) and average disease incidence (12.30 per cent) with average seed yield (1825 kg/ha) and test weight (16.02 gm). This was followed by Thiram and Carbendazim which recorded average disease incidence (13.20 per cent) and (15.60 per cent), disease control (80.19 per cent) and (76.59 per cent), average yield (1800

kg/ha) and (1625 kg/ha) and test weight (15.53 gm) and (15.50 gm). This was followed by *T. viride*, *T. harzianum* and *Pseudomonas fluroescens* were recorded average disease incidence (20.66 per cent) and (22.48 per cent) and (26.76 per cent), disease control (69.00per cent) and (66.27 per cent) and (59.85 per cent), average yield (1566 kg/ha) and (1466 kg/ha) and (1416 kg/ha) and test weight (15.12gm) and (14.79 gm) and (14.52 gm) respectively. Carbendazim + Mancozeb found least effective in disease incidence (30.20 per cent), disease control (54.69 per cent), average yield (1291 kg/ha) and test weight (14.36 gm) as compared to other treatments over control with disease incidence (66.66 per cent), disease control (00.00 per cent), average yield (1170 kg/ha) and test weight (11.67 gm).

Results obtained on varietal screening of Chickpea against *Macrophomina phaseolina* revealed varied reactions to dry root rot disease and exhibited the wide range of disease reaction.

Amongst 20 germplasm lines 10 lines viz., BCP-73, PG-12, Jaki-9218, BCP-3, PG-9425-5, BCP-107, PG-9557-2-1, PG-9758-6-2, BCP-15 and BCP-28 were found to be resistant with an average disease incidence (6.88 per cent), 4 lines viz., Vishal , Virat, PG-96329 and PG-9757-6-1 were found to be moderately resistant with average incidence (15.24 per cent), 2 lines viz., PG-94094 and PG-9755-8-1 were found to be susceptible with average disease incidence (29.03 per cent) and 4 lines viz., PG-97022, BDNG-2002, PG-97313, PG-98902, and were found to be highly susceptible with average disease incidence (78.28 per cent).

Conclusion

Thus, from the result obtained on various aspects during present investigations on dry root rot of Chickpea caused by *Macrophomina phaseolina* (Tassi) Goid can be concluded as under:

1. *Macrophomina phaseolina* (Tassi) Goid is one of the most destructive pathogen of Chickpea causing seed rot, root rot, stem rot, damping off and leaf blight disease syndromes.
2. Survey of Chickpea dry root rot disease in Latur district revealed that the average incidence of Chickpea dry root rot disease was observed to the tune of 13.85 per cent with maximum of 21.60 per cent in Ausa tahsil.
3. Studies on isolation, pathogenecity and symptomatology of Chickpea dry root rot disease revealed that the *Macrophomina phaseolina* (Tassi) Goid was the major cause of Chickpea dry root rot disease in the Latur district.
4. The pathogen (*Macrophamina phaseolina*) could grow and proliferated better on a wide range of agar media preferably on Potato dextrose agar, Host leaf extract agar, Czapek's dox agar, Richards's agar.
5. Pot culture experiment in chemical seed treatment Carbendazim was recorded highest seed germination (90.00 per cent) less disease incidence (14.00 per cent) and maximum disease control (86 per cent). In bioagent *T. viride* was recorded in highest seed germination (92.00 per cent), less disease incidence (31.15 per cent), and maximum disease control (68.85 per cent). In organic soil amendments Neem cake was recorded seed germination (85.66 per cent), disease incidence (14.70 per cent) and disease control (85.30 per cent).

6. In field condition chemicals seed treatment group fungicides Carbendazim + Thiram significantly reduced disease incidence and increase average seed yield and test weight. In biological control seed treatment with *T. harzianum* + *T. viride* significantly reduced disease incidence and increased average seed yield and test weight.

7. In screening of Chickpea cultivar, genotypes, germplasm lines were found resistant, moderately resistant, susceptible and highly susceptible for dry root rot disease. In varietal screening of Chickpea germplasm lines against *Macrophomina phaseolina* 10 germplasm lines were found to be resistant to *Macrophomina* dry root rot of Chickpea. These lines could be encouraged for further breeding programmes.

Thus Chickpea dry root rot incited by *Macrophomina phaseolina* (Tassi) Goid. Being a soil borne disease could be managed by the integrated management practices like seed treatment with chemicals, seed treatment with bioagents and soil application of organic soil amendments.

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RESEARCH TITLE: “STUDIES ON DRY ROOT ROT OF CHICKPEA (*Cicer arietinum* L.) CAUSED BY *Macrophomina phaseolina*. (Tassi) Goid ”.

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ABSTRACT

Chickpea (*Cicer arietinum* L.) is the world's third most important crop and also important in Maharashtra. Pathogen *Macrophomina phaseolina* causing dry root rot disease in Chickpea. It has been found to affect severely and causes accountable qualitative and quantitative losses in Chickpea which plays an important role in decreasing Chickpea production and productivity in a state. Present investigations on *Macrophomina phaseolina* were carried out during *rabi* 2010, on the aspect viz, survey on dry root rot, isolation, pathogenicity, cultural characteristic of pathogen *Macrophomina phaseolina*, *in vitro* evaluation of bioagents, fungicides and organic amendments (pot culture) against *Macrophomina phaseolina*, *in vivo* evaluation of fungicides and bioagents and germplasm screening at the department of Plant Pathology, College of Agriculture Latur.

Survey of Chickpea dry root rot disease in Latur district revealed that the average incidence of Chickpea dry root rot disease to the tune of 13.85 per cent and tahsilwise survey report indicate maximum disease incidence in in AUSA tahsil 21.60 per cent and minimum disease incidence in Nilanga tahsil (9.15 per cent). The test pathogen was successfully isolated on Potato Dextrose Agar (PDA) and proved its pathogenicity on Chickpea Cv. JG-62 under screen house conditions. Out of six culture media, potato dextrose agar was found most suitable which encouraged maximum radial mycelia growth and sclerotial production this was followed by host leaf extract agar, Czapek's dox agar, Richards's agar.

In pot culture experiment evaluation of seed treatments with fungicides, bioagents and organic soil amendments with Chickpea Cv.-62. Seed with treatment Carbendazim was recorded highest seed germination (90.00 per cent) least disease incidence (14.00 per cent) and maximum disease control (86 per cent). In bioagent *T. viride* was recorded highest seed germination (92.00 per cent), least disease incidence (31.15 per cent), and maximum disease control (68.85 per cent). In organic soil amendments Neem cake was recorded highest seed germination (85.66 per cent), least disease incidence (14.70 per cent) and maximum disease control (85.30 per cent).

In field evaluation of fungicides and bioagents with Chickpea Cv. JG-62. In chemical control seed treatment with fungicides Carbendazim + Thiram significantly reduced disease incidence (10.86 per cent) with maximum disease controls (83.70 per cent) with highest seed yield (1950 kg/ha) and test weight (16.36 gm). In biological control seed treatment with *T. harzianum* + *T. viride* significantly reduced disease incidence (12.30 per cent) with maximum disease control (81.54 per cent) with highest average seed yield (1825 kg/ha) and test weight (16.02 gm).

In pot culture experiment Chickpea germplasm screening against *Macrophomina phaseolina* causing dry root rot disease in Chickpea, 20 germplasm lines were screened 10 lines were found to be resistant, 4 lines were found to be moderately resistant, 2 lines were found to be susceptible and 4 lines were found to be highly susceptible respectively.